

ggNetView manual documentation

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Chapter 1

ggNetView manual documentation

1.1 Introduction

Network analysis has been widely applied in the **life sciences, microbiology, ecology, and agronomy** to quantify interactions among **genes, proteins, metabolites, microorganisms, and environmental factors**. These interactions collectively shape the functioning of biological and ecological systems. Despite its broad adoption, existing tools often face limitations in flexibility, customization, reproducibility, and the generation of publication-ready figures.

To address these challenges, **ggNetView** was developed as an **R package** that integrates **ggplot2, ggraph, and tidygraph** within the **Grammar of Graphics framework**, enabling fully reproducible and highly customizable network visualizations. The package provides deterministic layout algorithms, comprehensive topological analyses, modular coloring schemes, hierarchical annotations, and consistent theming, ensuring standardized and interpretable graphical output.

Although initially designed for applications in **soil science and microbial ecology**, **ggNetView** is broadly applicable to network analyses in **molecular biology**, including **WGCNA and protein–protein interaction (PPI) networks**. By lowering technical barriers in network construction and visualization, **ggNetView** enables researchers across disciplines to efficiently produce reproducible, publication-quality network figures.

1.2 Installation

Step1: install CRAN dependencies

```
cran_pkgs <- c(
  "boot", "dplyr", "FNN", "future", "future.apply",
  "ggforce", "ggnewscale", "ggplot2", "gggraph", "ggrepel",
  "Hmisc", "huge", "igraph", "magrittr", "MASS",
  "Matrix", "patchwork", "progressr", "psych", "purrr",
  "qgraph", "Rcpp", "RcppArmadillo", "readr", "rlang",
  "scales", "scatterpie", "stringr", "tibble", "tidygraph",
  "tidyr", "vegan", "VGAM", "WGCNA"
)

new_pkgs <- cran_pkgs[!cran_pkgs %in% installed.packages()[, "Package"]]
if (length(new_pkgs)) install.packages(new_pkgs)
```

Step2: (optional) install suggested packages

These packages are not required for the core functionality, but enable additional features (e.g. dynamic tree cut, node influence, vignettes, tests):

```
install.packages(c("dynamicTreeCut", "influential",
  "knitr", "rmarkdown",
  "RobustRankAggreg", "testthat"))
```

Step3: install ggNetView from GitHub

```
# install.packages("devtools")
devtools::install_github("Jiawang1209/ggNetView")
```

1.3 Citation

If you use ggNetView in your research, please cite:

Yue Liu, Chao Wang (2025). ggNetView: An R package for complex biological and ecological
<https://github.com/Jiawang1209/ggNetView>

1.4 Source Code

The source code for ggNetView is available in the ggNetView repository.

<https://github.com/Jiawang1209/ggNetView>

1.5 Contact

- Email: Jiawang1209@163.com
- Github: <https://github.com/Jiawang1209/ggNetView>
- ggNetView-manual: <https://jiawang1209.github.io/ggNetView-manual/>

Chapter 2

Build graph object

Basic workflow of **ggNetView**

1. Build a **graph object**
 2. Understand and manipulate the **graph object**
 3. Visualize the network using **layout algorithms**
 4. Retrieve **topology** of network
-

Load R Package

```
library(tidyverse)
library(ggNetView)
library(igraph)
```

2.1 Build graph from matrix

Example data

```
# Access built-in example datasets in ggNetView

# Raw ASV/OTU abundance table (rows = ASVs/OTUs, cols = samples)
data("otu_tab")
dim(otu_tab)
```

```
## [1] 2859 18
```

```
otu_tab[1:5, 1:5]
```

```
##           K01  K02  K03  K04  K05
## ASV_1 1113 1968  816 1372 1062
## ASV_2 1922 1227 2355 2218 2885
## ASV_3  568  460  899  902 1226
## ASV_4 1433  400  535  759 1287
## ASV_6  882  673  819  888 1475
```

```
# Rarefied ASV/OTU table (sequencing depth normalised by rarefaction)
data("otu_rare")
dim(otu_rare)
```

```
## [1] 2859 18
```

```
otu_rare[1:5, 1:5]
```

```
##           K01  K02  K03  K04  K05
## ASV_1  992 1636  604 1084  806
## ASV_2 1725 1018 1814 1743 2196
## ASV_3  520  389  687  701  932
## ASV_4 1280  328  425  580 1004
## ASV_6  794  557  633  706 1142
```

```
# Rarefied + converted to relative abundance - most common downstream input
data("otu_rare_relative")
dim(otu_rare_relative)
```

```
## [1] 2859 18
```

```
otu_rare_relative[1:5, 1:5]
```

```
##           K01           K02           K03           K04           K05
## ASV_1 0.03306667 0.05453333 0.02013333 0.03613333 0.02686667
## ASV_2 0.05750000 0.03393333 0.06046667 0.05810000 0.07320000
## ASV_3 0.01733333 0.01296667 0.02290000 0.02336667 0.03106667
## ASV_4 0.04266667 0.01093333 0.01416667 0.01933333 0.03346667
## ASV_6 0.02646667 0.01856667 0.02110000 0.02353333 0.03806667
```

```
# Taxonomic annotation for the ASVs/OTUs above.
# First column must be the ASV/OTU ID (matching rownames of the abundance tables).
data("tax_tab")
dim(tax_tab)
```

```
## [1] 2859      8
```

```
tax_tab[1:5, 1:5]
```

```
## # A tibble: 5 x 5
##   OTUID   Kingdom Phylum      Class      Order
##   <chr>   <chr>    <chr>    <chr>    <chr>
## 1 ASV_2   Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 2 ASV_3   Bacteria Verrucomicrobia  Spartobacteria  Unassigned
## 3 ASV_31  Bacteria Actinobacteria  Actinobacteria  Actinomycetales
## 4 ASV_27  Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 5 ASV_9   Bacteria Unassigned      Unassigned  Unassigned
```

Build graph object

```
# Build a co-occurrence network directly from an abundance matrix.
# Internally: (1) optional transform -> (2) correlation -> (3) p-value adjustment
# -> (4) edge filtering -> (5) community detection -> (6) attach taxonomy.
graph_obj <- build_graph_from_mat(
  mat           = otu_rare_relative, # variables x samples numeric matrix
  transform.method = "none",          # input is already relative abundance
  r.threshold    = 0.7,               # |r| cutoff (subjective; consider ggNetView_RMT)
  p.threshold    = 0.05,              # adjusted p-value cutoff
  method         = "WGCNA",          # correlation backend: WGCNA::corAndPvalue
  cor.method     = "pearson",         # Pearson correlation
  proc           = "bonferroni",      # multiple-testing correction
  module.method  = "Fast_greedy",     # community detection algorithm
  node_annotation = tax_tab,          # taxonomy joined onto each node by name
  top_modules    = 15,               # keep top-15 modules; rest -> "Others"
  seed           = 1115              # reproducibility
)

# Inspect node/edge counts and the columns now available on nodes:
# modularity, modularity2, modularity3, Modularity, Degree, Strength + taxonomy
graph_obj
```

```
## # A tbl_graph: 213 nodes and 844 edges
## #
```

```
## # An undirected simple graph with 29 components
## #
## # Node Data: 213 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_649 5          5          5          5          27      26.5 Bacter~
## 2 ASV_705 5          5          5          5          27      26.5 Bacter~
## 3 ASV_12~ 5          5          5          5          27      26.5 Bacter~
## 4 ASV_13~ 5          5          5          5          27      26.5 Bacter~
## 5 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 6 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 7 ASV_24~ 5          5          5          5          27      26.5 Bacter~
## 8 ASV_25~ 5          5          5          5          27      26.4 Bacter~
## 9 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## 10 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## # i 203 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 844 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1 194    195 0.959      0.959 Positive
## 2 185    208 0.954      0.954 Positive
## 3 185    213 0.957      0.957 Positive
## # i 841 more rows
```

2.2 Build graph from data frame

Example data

```
# Edge-list style example data (protein-protein interactions).
data("ppi_example")

# `df`: edge list - typically 3 columns: node1 | node2 | weight (optional)
df = ppi_example$ppi
head(df)

##   from to    weight
## 1  A1 D40 9.306533
## 2  A2 D39 11.783920
```

```
## 3   A3 D38 23.005025
## 4   A4 D37  7.412060
## 5   A5 D36 18.778894
## 6   A6 D35 16.592965
```

```
# `node_annotation`: per-node metadata; first column = node ID
node_annotation = ppi_example$annotation
head(node_annotation)
```

```
##   node group
## 1   A1      A
## 2   A2      A
## 3   A3      A
## 4   A4      A
## 5   A5      A
## 6   A6      A
```

Build graph object

```
# Build a tbl_graph from an edge-list data.frame (no correlation step needed -
# edges are taken as-is from `df`). Community detection still runs on the result.
graph_obj_from_df <- build_graph_from_df(
  df           = df,           # edge list
  node_annotation = node_annotation, # per-node metadata (joined by name)
  directed      = F,           # undirected network
  module.method = "Fast_greedy", # community detection
  top_modules   = 15,           # keep top-15 modules; rest -> "Others"
  seed          = 1115
)

graph_obj_from_df
```

```
## # A tbl_graph: 100 nodes and 50 edges
## #
## # An unrooted forest with 50 trees
## #
## # Node Data: 100 x 9 (active)
```

	name	group	modularity	modularity2	modularity3	Modularity	Degree	Segree
	<chr>	<chr>	<fct>	<fct>	<chr>	<fct>	<dbl>	<dbl>
## 1	C13	C	1	1	1	1	1	1
## 2	C28	C	1	1	1	1	1	1
## 3	C2	C	10	10	10	10	1	1
## 4	D9	D	10	10	10	10	1	1
## 5	A3	A	11	11	11	11	1	1

```
## 6 D38 D 11 11 11 11 1 1
## 7 B12 B 12 12 12 12 1 1
## 8 D19 D 12 12 12 12 1 1
## 9 A1 A 13 13 13 13 1 1
## 10 D40 D 13 13 13 13 1 1
## # i 90 more rows
## # i 1 more variable: Strength <dbl>
## #
## # Edge Data: 50 x 4
## from to weight correlation
## <int> <int> <dbl> <dbl>
## 1 9 10 45.2 45.2
## 2 15 16 50.6 50.6
## 3 5 6 37.8 37.8
## # i 47 more rows
```

```
# Same as above, but without any node annotation table.
# Useful when you only have an edge list and no extra metadata to attach.
graph_obj_from_df2 <- build_graph_from_df(
  df = df,
  node_annotation = NULL, # skip the annotation join entirely
  directed = F,
  module.method = "Fast_greedy",
  top_modules = 15,
  seed = 1115
)

graph_obj_from_df2
```

```
## # A tbl_graph: 100 nodes and 50 edges
## #
## # An unrooted forest with 50 trees
## #
## # Node Data: 100 x 8 (active)
## name modularity modularity2 modularity3 Modularity Degree Segree Strength
## <chr> <fct> <fct> <chr> <fct> <dbl> <dbl> <dbl>
## 1 C13 1 1 1 1 1 1 26.7
## 2 C28 1 1 1 1 1 1 26.7
## 3 C2 10 10 10 10 1 1 37.4
## 4 D9 10 10 10 10 1 1 37.4
## 5 A3 11 11 11 11 1 1 37.8
## 6 D38 11 11 11 11 1 1 37.8
## 7 B12 12 12 12 12 1 1 41.3
## 8 D19 12 12 12 12 1 1 41.3
## 9 A1 13 13 13 13 1 1 45.2
```



```
## 10 D40    13          13          13          13          1    1    45.2
## # i 90 more rows
## #
## # Edge Data: 50 x 4
##   from    to weight correlation
##   <int> <int> <dbl>         <dbl>
## 1     9    10  45.2          45.2
## 2    15    16  50.6          50.6
## 3     5     6  37.8          37.8
## # i 47 more rows
```

2.3 Build graph from data frame with module

Example data

```
# Like ppi_example, but `annotation` already contains a `Modularity` column
# (i.e. module assignments computed elsewhere - we will NOT recompute them).
data("ppi_module")
```

```
df = ppi_module$ppi
head(df)
```

```
##   from to   weight
## 1  A1 D40 9.306533
## 2  A2 D39 11.783920
## 3  A3 D38 23.005025
## 4  A4 D37  7.412060
## 5  A5 D36 18.778894
## 6  A6 D35 16.592965
```

```
node_annotation = ppi_module$annotation
head(node_annotation) # note: must contain a `Modularity` column
```

```
##   node Modularity
## 1   A1          A
## 2   A2          A
## 3   A3          A
## 4   A4          A
## 5   A5          A
## 6   A6          A
```

Build graph object

```

# When you already have pre-computed module assignments and want to keep them
# (e.g. from a previous WGCNA run, expert curation, biological prior),
# use the `_module` variant - it skips internal community detection and uses
# the Modularity column from `node_annotation` directly.
graph_obj_from_module <- build_graph_from_module(
  df          = df,
  node_annotation = node_annotation,  # MUST contain `Modularity` column
  directed     = F,
  top_modules  = 15,
  seed        = 1115
)

graph_obj_from_module

```

```

## # A tbl_graph: 100 nodes and 50 edges
## #
## # An unrooted forest with 50 trees
## #
## # Node Data: 100 x 7 (active)
##   name  Modularity modularity2 modularity3 Degree Segree Strength
##   <chr> <ord>      <ord>        <chr>      <dbl>  <dbl>  <dbl>
## 1 D1    D          D          D          1      1      37.1
## 2 D2    D          D          D          1      1      63.9
## 3 D3    D          D          D          1      1      54.6
## 4 D4    D          D          D          1      1      61.7
## 5 D5    D          D          D          1      1      36.1
## 6 D6    D          D          D          1      1      71.0
## 7 D7    D          D          D          1      1      27.9
## 8 D8    D          D          D          1      1      34.6
## 9 D9    D          D          D          1      1      37.4
## 10 D10  D          D          D          1      1      63.2
## # i 90 more rows
## #
## # Edge Data: 50 x 4
##   from  to weight correlation
##   <int> <int> <dbl>    <dbl>
## 1    40   91  45.2      45.2
## 2    39   92  50.6      50.6
## 3    38   93  37.8      37.8
## # i 47 more rows

```

2.4 Build graph from adjacency matrix

Example data

```
# Square adjacency matrix (already-filtered weights for every node pair).
# Useful when the network was built externally (NetCoMi, FlashWeave, ...).
data("adjacency_matrix_example")
dim(adjacency_matrix_example)
```

```
## [1] 2859 2859
```

```
adjacency_matrix_example[1:5, 1:5]
```

```
##      ASV_1      ASV_2 ASV_3 ASV_4      ASV_6
## ASV_1      0 0.0000000      0      0 0.0000000
## ASV_2      0 0.0000000      0      0 0.8947427
## ASV_3      0 0.0000000      0      0 0.0000000
## ASV_4      0 0.0000000      0      0 0.0000000
## ASV_6      0 0.8947427      0      0 0.0000000
```

```
# Reuse the same taxonomy table as node annotation.
data("tax_tab")
dim(tax_tab)
```

```
## [1] 2859      8
```

```
tax_tab[1:5, 1:5]
```

```
## # A tibble: 5 x 5
##   OTUID Kingdom Phylum      Class      Order
##   <chr>   <chr>   <chr>      <chr>      <chr>
## 1 ASV_2  Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 2 ASV_3  Bacteria Verrucomicrobia  Spartobacteria  Unassigned
## 3 ASV_31 Bacteria Actinobacteria  Actinobacteria  Actinomycetales
## 4 ASV_27 Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 5 ASV_9  Bacteria Unassigned      Unassigned      Unassigned
```

Build graph object

```

# Build a tbl_graph from a precomputed adjacency matrix.
# Community detection is run internally (Fast_greedy here) because we have
# weights but no module assignments yet.
graph_obj_adj <- build_graph_from_adj_mat(
  adjacency_matrix = adjacency_matrix_example,
  module.method    = "Fast_greedy",
  node_annotation  = tax_tab,
  top_modules      = 15,
  seed             = 1115
)

graph_obj_adj

```

```

## # A tbl_graph: 2049 nodes and 9602 edges
## #
## # An undirected simple graph with 100 components
## #
## # Node Data: 2,049 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <dbl>      <dbl>      <dbl>      <dbl>      <dbl>  <dbl>  <chr>
## 1 ASV_916 1          1          1          1          58      50.5 Bacter~
## 2 ASV_777 1          1          1          1          58      48.7 Bacter~
## 3 ASV_606 1          1          1          1          55      45.8 Bacter~
## 4 ASV_740 1          1          1          1          54      47.2 Bacter~
## 5 ASV_14~ 1          1          1          1          54      44.5 Bacter~
## 6 ASV_23~ 1          1          1          1          54      47.4 Bacter~
## 7 ASV_15~ 1          1          1          1          52      45.3 Bacter~
## 8 ASV_24~ 1          1          1          1          52      43.0 Bacter~
## 9 ASV_19~ 1          1          1          1          52      43.0 Bacter~
## 10 ASV_568 1          1          1          1          51      45.1 Bacter~
## # i 2,039 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 9,602 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1 1771 1825 0.793      0.793 Positive
## 2 594 597 0.895      0.895 Positive
## 3 588 597 0.864      0.864 Positive
## # i 9,599 more rows

```

2.5 Build graph from adjacency matrix with module information

```
# Add a pre-existing module assignment to the annotation table.
# Here we just (for demonstration) treat the Kingdom column as the module label;
# in real use, this should be your actual module assignments.
```

```
tax_tab_module <- tax_tab %>%
  dplyr::mutate(Modularity = Kingdom)
```

```
# `_module` variant for adjacency input: REQUIRES `node_annotation` to contain
# a `Modularity` column. Skips community detection and uses your provided
# module assignments directly.
```

```
graph_obj_adj_module <- build_graph_from_adj_mat_module(
  adjacency_matrix = adjacency_matrix_example,
  node_annotation = tax_tab_module,    # MUST contain `Modularity` column
  top_modules     = 15,
  seed            = 1115
)
```

```
## The max module in network is 3 we use the 3 modules for next analysis
```

```
graph_obj_adj_module
```

```
## # A tbl_graph: 2049 nodes and 9602 edges
```

```
## #
```

```
## # An undirected simple graph with 100 components
```

```
## #
```

```
## # Node Data: 2,049 x 14 (active)
```

	name	Kingdom	Phylum	Class	Order	Family	Genus	Species	Modularity	modularity2
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<ord>	<ord>
## 1	ASV_9~	Bacter~	Prote~	Gamm~	Unas~	Unass~	Unas~	Unassi~	Bacteria	Bacteria
## 2	ASV_7~	Bacter~	Acido~	Acid~	Unas~	Unass~	Gp6	Unassi~	Bacteria	Bacteria
## 3	ASV_6~	Bacter~	Actin~	Acti~	Acti~	Nocar~	Marm~	Unassi~	Bacteria	Bacteria
## 4	ASV_3~	Bacter~	Prote~	Beta~	Unas~	Unass~	Unas~	Unassi~	Bacteria	Bacteria
## 5	ASV_7~	Bacter~	Acido~	Acid~	Unas~	Unass~	Gp16	Unassi~	Bacteria	Bacteria
## 6	ASV_1~	Bacter~	Actin~	Acti~	Unas~	Unass~	Unas~	Unassi~	Bacteria	Bacteria
## 7	ASV_2~	Bacter~	Prote~	Alph~	Rhiz~	Hyphe~	Rhod~	Unassi~	Bacteria	Bacteria
## 8	ASV_25	Bacter~	Verru~	Spar~	Unas~	Unass~	Spar~	Unassi~	Bacteria	Bacteria
## 9	ASV_7~	Bacter~	Actin~	Acti~	Acti~	Cellu~	Cell~	Cellul~	Bacteria	Bacteria
## 10	ASV_1~	Bacter~	Acido~	Acid~	Unas~	Unass~	Gp3	Unassi~	Bacteria	Bacteria

```
## # i 2,039 more rows
```

```
## # i 4 more variables: modularity3 <chr>, Degree <dbl>, Segree <dbl>,
```

```
## # Strength <dbl>
## #
## # Edge Data: 9,602 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>         <dbl> <chr>
## 1  1211  1492  0.793          0.793 Positive
## 2   323  2013  0.895          0.895 Positive
## 3   267  2013  0.864          0.864 Positive
## # i 9,599 more rows
```

2.6 Build graph from double matrix

Example data

```
# Two abundance matrices on the SAME samples but DIFFERENT feature sets,
# typically used for cross-kingdom networks (e.g. bacteria <-> fungi).
```

```
# Bacterial ASV table
data("BASV_tab")
dim(BASV_tab)
```

```
## [1] 50 10
```

```
BASV_tab[1:5, 1:5]
```

```
##           Sample1 Sample2 Sample3 Sample4 Sample5
## BASV1  7.769249 14.676091  6.822246 11.806957 12.086254
## BASV2 10.771846  7.749385 14.966987  7.486507 12.645461
## BASV3 11.746937 10.023547 10.025906 11.967562  8.748440
## BASV4  8.210859  4.408296 10.070862  8.954202 14.639298
## BASV5  9.833271  9.776280 10.465463  8.380012  8.905738
```

```
# Fungal ASV table
data("FASV_tab")
dim(FASV_tab)
```

```
## [1] 50 10
```

```
FASV_tab[1:5, 1:5]
```

```
##           Sample1  Sample2  Sample3  Sample4  Sample5
## FASV1  8.195641  7.715472 13.247401 10.419618 11.961410
## FASV2 10.032701 12.427388  7.693471 10.711077 12.157527
## FASV3 10.109549 11.891917  9.607849  8.256637  9.989264
## FASV4  9.817056  8.038730  5.519155  7.601228 12.688531
## FASV5 13.370263  9.795554 13.639041 11.071635  8.176218
```

```
# Combined node annotation table covering BOTH feature sets above.
data("double_mat_node_df")
dim(double_mat_node_df)
```

```
## [1] 100  2
```

```
head(double_mat_node_df)
```

```
##   name      type
## 1 BASV1 Bacterial
## 2 BASV2 Bacterial
## 3 BASV3 Bacterial
## 4 BASV4 Bacterial
## 5 BASV5 Bacterial
## 6 BASV6 Bacterial
```

Build graph object

```
# Cross-domain bipartite-style network:
# correlations are computed BETWEEN features in mat1 and features in mat2
# (no within-mat1 / within-mat2 edges), then thresholded, modularised, joined
# with node annotation.
graph_obj_double_mat <- build_graph_from_double_mat(
  mat1      = BASV_tab,      # set A (e.g. bacteria)
  mat2      = FASV_tab,      # set B (e.g. fungi)
  module.method = "Fast_greedy",
  node_annotation = double_mat_node_df,
  top_modules   = 15,
  seed          = 1115
)
```

```
## The max module in network is 4 we use the 4 modules for next analysis
```

```
graph_obj_double_mat
```

```
## # A tbl_graph: 100 nodes and 2500 edges
## #
## # A bipartite simple graph with 1 component
## #
## # Node Data: 100 x 9 (active)
##   name      type      modularity modularity2 modularity3 Modularity Degree Segree
##   <chr>    <chr>    <fct>      <fct>      <chr>      <fct>      <dbl>  <dbl>
## 1 BASV3   Bacterial 1          1          1          1          50    50
## 2 BASV6   Bacterial 1          1          1          1          50    50
## 3 BASV8   Bacterial 1          1          1          1          50    50
## 4 BASV13  Bacterial 1          1          1          1          50    50
## 5 BASV17  Bacterial 1          1          1          1          50    50
## 6 BASV19  Bacterial 1          1          1          1          50    50
## 7 BASV25  Bacterial 1          1          1          1          50    50
## 8 BASV26  Bacterial 1          1          1          1          50    50
## 9 BASV27  Bacterial 1          1          1          1          50    50
## 10 BASV31 Bacterial 1          1          1          1          50    50
## # i 90 more rows
## # i 1 more variable: Strength <dbl>
## #
## # Edge Data: 2,500 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1    15    57  0.338      -0.338 Negative
## 2    57    90  0.648       0.648 Positive
## 3    16    57  0.162       0.162 Positive
## # i 2,497 more rows
```

2.7 Build graph from igraph object

```
# Load the PPI example dataset (edge list + node annotation).
data("ppi_example")

# First build a native igraph object directly from the edge list.
# This is what users typically already have when they come from an igraph workflow.
igraph_object <- igraph::graph_from_data_frame(
  d      = ppi_example$ppi,          # edge list
  vertices = ppi_example$annotation, # vertex metadata
  directed = FALSE
)
```



```
igraph_object
```

```
## IGRAPH 4da318a UNW- 100 200 --
## + attr: name (v/c), group (v/c), weight (e/n)
## + edges from 4da318a (vertex names):
## [1] A1 --D40 A2 --D39 A3 --D38 A4 --D37 A5 --D36 A6 --D35 A7 --D34 A8 --D33
## [9] A9 --D32 A10--D31 B1 --D30 B2 --D29 B3 --D28 B4 --D27 B5 --D26 B6 --D25
## [17] B7 --D24 B8 --D23 B9 --D22 B10--D21 B11--D20 B12--D19 B13--D18 B14--D17
## [25] B15--D16 B16--D15 B17--D14 B18--D13 B19--D12 B20--D11 C1 --D10 C2 --D9
## [33] C3 --D8 C4 --D7 C5 --D6 C6 --D5 C7 --D4 C8 --D3 C9 --D2 C10--D1
## [41] C11--C30 C12--C29 C13--C28 C14--C27 C15--C26 C16--C25 C17--C24 C18--C23
## [49] C19--C22 C20--C21 C20--C21 C19--C22 C18--C23 C17--C24 C16--C25 C15--C26
## [57] C14--C27 C13--C28 C12--C29 C11--C30 C10--D1 C9 --D2 C8 --D3 C7 --D4
## + ... omitted several edges
```

```
# Convert an existing igraph object into a ggNetView-compatible tbl_graph.
# Adds Modularity / Degree / Strength columns and harmonises the schema so it
# can be plugged straight into ggNetView().
```

```
graph_obj_from_igraph <- build_graph_from_igraph(
  igraph      = igraph_object,
  module.method = "Fast_greedy",
  seed        = 1115
)
```

```
graph_obj_from_igraph
```

```
## # A tbl_graph: 100 nodes and 50 edges
```

```
## #
```

```
## # An unrooted forest with 50 trees
```

```
## #
```

```
## # Node Data: 100 x 8 (active)
```

	name	group	modularity	modularity2	modularity3	Modularity	Degree	Strength
	<chr>	<chr>	<fct>	<ord>	<chr>	<ord>	<dbl>	<dbl>
## 1	C13	C	1	1	1	1	1	26.7
## 2	C28	C	1	1	1	1	1	26.7
## 3	C2	C	10	10	10	10	1	37.4
## 4	D9	D	10	10	10	10	1	37.4
## 5	A3	A	11	11	11	11	1	37.8
## 6	D38	D	11	11	11	11	1	37.8
## 7	B12	B	12	12	12	12	1	41.3
## 8	D19	D	12	12	12	12	1	41.3
## 9	A1	A	13	13	13	13	1	45.2
## 10	D40	D	13	13	13	13	1	45.2

```
## # i 90 more rows
```

```
## #
## # Edge Data: 50 x 3
##   from    to weight
##   <int> <int> <dbl>
## 1     9    10  45.2
## 2    15    16  50.6
## 3     5     6  37.8
## # i 47 more rows
```

2.8 Build graph from WGCNA

```
# Workflow for importing an existing WGCNA pipeline output.
# (Not evaluated here because it requires external files saved by WGCNA::blockwiseModu

# 1. Load the topological overlap matrix (TOM) saved by WGCNA.
load(file = "WGCNA.tom-block.1.RData")
TOM_mat <- as.matrix(TOM)

# 2. Load the abundance matrix used to build that TOM (for any downstream
#     re-thresholding / sanity checks).
plot.mat = readRDS(file = "plot.mat.Rds")

# 3. Sparsify the TOM: keep only the top_k strongest TOM connections per node.
#     `trans_TOM_in_WGCNA()` is a helper bundled with ggNetView for this step.
trans_TOM <- trans_TOM_in_WGCNA(TOM_mat, plot.mat, top_k = 1)

# 4. Load the WGCNA net object (contains module colour assignments).
net <- readRDS(file = "net.Rds")

# 5. Build the module table - one row per feature, columns: ID + Module.
module <- data.frame(
  ID      = names(net$colors),
  Module  = as.character(net$colors)
)

# 6. Optional ordering of node IDs (used as node_annotation here).
id <- data.frame(
  ID = names(sort(net$colors))
)

# 7. Assemble the tbl_graph: uses the sparsified TOM as edge weights and the
#     WGCNA module colours as the Modularity assignment (no recomputation).
obj <- build_graph_from_wgcna(
  wgcna_tom      = trans_TOM,
```

```

    module          = module,
    node_annotation = id
  )

obj

```

2.9 Build graph from multi-graph

```

# Consensus network workflow:
# (1) Build per-method networks on the SAME matrix.
# (2) Convert each tbl_graph back to an adjacency matrix.
# (3) Fuse the per-method adjacencies into ONE consensus tbl_graph.

# Restrict to the 40 most abundant OTUs to keep this chunk responsive.
data(otu_rare_relative)
mat <- as.matrix(otu_rare_relative)
mat <- mat[order(rowSums(mat), decreasing = TRUE)[seq_len(40)], ]

# ---- (1) Three per-method networks on the same data ----
# Spearman correlation through psych::corr.test - baseline.
g_cor <- build_graph_from_mat(mat, method = "cor", cor.method = "spearman",
                             proc = "BH", r.threshold = 0.4, p.threshold = 0.05,
                             module.method = "Fast_greedy", seed = 1)

## The max module in network is 5 we use the 5 modules for next analysis

# WGCNA-style correlation (WGCNA::corAndPvalue + the package's thresholding).
g_wgcna <- build_graph_from_mat(mat, method = "WGCNA", cor.method = "spearman",
                              proc = "BH", r.threshold = 0.4, p.threshold = 0.05,
                              module.method = "Fast_greedy", seed = 1)

## The max module in network is 6 we use the 6 modules for next analysis

# Hmisc::rcorr - yet another correlation engine, slightly different tie/NA behaviour.
g_hmisc <- build_graph_from_mat(mat, method = "Hmisc", cor.method = "spearman",
                              r.threshold = 0.4, p.threshold = 0.05,
                              module.method = "Fast_greedy", seed = 1)

## The max module in network is 4 we use the 4 modules for next analysis

```

```

# ---- (2) tbl_graph -> adjacency matrix ----
# `get_graph_adjacency()` recovers the weighted adjacency matrix from a tbl_graph,
# which is the format expected by build_graph_from_consensus().
adj_cor    <- get_graph_adjacency(g_cor)
adj_wgcna  <- get_graph_adjacency(g_wgcna)
adj_hmisc  <- get_graph_adjacency(g_hmisc)

# Bundle the per-method matrices into a named list. The names ("cor", "wgcna",
# "hmisc") are kept as method labels in subsequent reports.
adj_list <- list(cor = adj_cor, wgcna = adj_wgcna, hmisc = adj_hmisc)

# ---- (3) Consensus fusion ----
# Borda rank fusion: ranks every pair by |w| in each method, averages the ranks,
# then min-max rescales to [-1, 1]. Robust to across-method weight-scale differences.
g_borda <- build_graph_from_consensus(
  adj_list          = adj_list,
  method            = "rank_fusion",
  rank_fusion_algorithm = "borda",
  threshold         = 0.5,          # final |w| cutoff applied to the fused matrix
  top_modules       = 5,
  seed              = 1
)

```

```
## The max module in network is 3 we use the 3 modules for next analysis
```

```
g_borda
```

```

## # A tbl_graph: 8 nodes and 8 edges
## #
## # An undirected simple graph with 3 components
## #
## # Node Data: 8 x 7 (active)
##   name    modularity modularity2 modularity3 Modularity Degree Strength
##   <chr>   <fct>      <ord>      <chr>      <ord>      <dbl>   <dbl>
## 1 ASV_6   1          1          1          1          3       2.33
## 2 ASV_10  1          1          1          1          3       2.33
## 3 ASV_2   1          1          1          1          3       2.33
## 4 ASV_17  1          1          1          1          3       2.33
## 5 ASV_36  2          2          2          2          1       1
## 6 ASV_38  2          2          2          2          1       1
## 7 ASV_33  3          3          3          3          1       1
## 8 ASV_37  3          3          3          3          1       1
## #
## # Edge Data: 8 x 5

```

```
##      from      to weight correlation corr_direction
##    <int> <int>  <dbl>          <dbl> <chr>
## 1      1      2  0.667          0.667 Positive
## 2      1      3  0.667          0.667 Positive
## 3      1      4  1            1      Positive
## # i 5 more rows
```

Chapter 3

Random Matrix Theory (RMT)–based random network

When a numerical matrix is used as input, it is necessary to filter and screen the resulting correlation matrix. Empirical thresholds are commonly adopted, usually between 0.6 to 0.85; however, these thresholds are largely subjective.

Compared with empirical thresholding, random matrix theory (RMT) provides a data-driven and objective approach for determining correlation thresholds, effectively separating true biological signals from random noise and improving the robustness and comparability of ecological networks.

3.1 Random Matrix Theory (RMT)

Load R Package

```
library(tidyverse)
library(ggNetView)
```

3.2 Example data

Example data

```
# Access built-in example datasets in ggNetView
```

```
# Relative abundance table of rarefied ASVs or OTUs
```

```
data("otu_rare_relative")
```

```
dim(otu_rare_relative)
```

```
## [1] 2859 18
```

```
otu_rare_relative[1:5, 1:5]
```

```
##           K01           K02           K03           K04           K05
## ASV_1 0.03306667 0.05453333 0.02013333 0.03613333 0.02686667
## ASV_2 0.05750000 0.03393333 0.06046667 0.05810000 0.07320000
## ASV_3 0.01733333 0.01296667 0.02290000 0.02336667 0.03106667
## ASV_4 0.04266667 0.01093333 0.01416667 0.01933333 0.03346667
## ASV_6 0.02646667 0.01856667 0.02110000 0.02353333 0.03806667
```

```
# Taxonomic annotation table of ASVs or OTUs
```

```
data("tax_tab")
```

```
dim(tax_tab)
```

```
## [1] 2859 8
```

```
tax_tab[1:5, 1:5]
```

```
## # A tibble: 5 x 5
```

```
##   OTUID Kingdom Phylum      Class      Order
##   <chr>  <chr>    <chr>      <chr>      <chr>
## 1 ASV_2  Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 2 ASV_3  Bacteria Verrucomicrobia Spartobacteria Unassigned
## 3 ASV_31 Bacteria Actinobacteria Actinobacteria Actinomycetales
## 4 ASV_27 Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 5 ASV_9  Bacteria Unassigned    Unassigned  Unassigned
```

3.3 Compute RMT threshold


```

# ---- RMT threshold scan #1: WGCNA correlation backend ----
# `ggNetView_RMT()` scans a series of candidate |r| thresholds, builds the
# thresholded correlation matrix at each, computes its eigenvalue spacing
# distribution (NNSD), and picks the threshold at which the NNSD transitions
# from Gaussian Orthogonal Ensemble (signal + noise) to Poisson (noise-free,
# modular network) - this is the RMT-recommended cutoff.
out <- ggNetView_RMT(
  mat           = otu_rare_relative, # input matrix (variables x samples)
  transfrom.method = "none",          # already relative-abundance, no extra transform
  method        = "WGCNA",           # use WGCNA::corAndPvalue for correlation
  cor.method     = "pearson",         # Pearson correlation
  unfold.method  = "gaussian",        # Gaussian KDE unfolding of eigenvalues
  bandwidth      = "nrd0",           # KDE bandwidth rule (stats::density default)
  nr.fit.points  = 20,                # support points (only used if unfold.method = "spline")
  discard.outliers = TRUE,            # IQR-trim extreme eigenvalues before unfolding
  discard.zeros  = TRUE,             # drop all-zero rows/cols after thresholding
  min.mat.dim    = 40,                # early-stop if effective dim drops below 40
  max.ev.spacing = 3,                 # truncate NNSD tail at spacing = 3
  save_plots     = FALSE,             # don't write diagnostic PNGs to disk
  out_dir        = "RMT_plots",       # (only used when save_plots = TRUE)
  verbose        = TRUE,              # print scan progress to console
  seed           = 1115               # fix RNG for reproducibility
)

```

```
## [Info] Matrix dimension: 2859 x 2859
```

```
## [Info] #non-zeros: 8173751 | Sparseness: 0.0000
```

```
## [Scan] 1/51 | threshold = 0
```

```
## Effective dimension: 2859
```

```
## [Scan] 2/51 | threshold = 0.02
```

```
## Effective dimension: 2859
```

```
## [Scan] 3/51 | threshold = 0.04
```

```
## Effective dimension: 2859
```

```
## [Scan] 4/51 | threshold = 0.06
```

```

##           Effective dimension: 2859

## [Scan]    5/51 | threshold = 0.08

##           Effective dimension: 2859

## [Scan]    6/51 | threshold = 0.1

##           Effective dimension: 2859

## [Scan]    7/51 | threshold = 0.12

##           Effective dimension: 2859

## [Scan]    8/51 | threshold = 0.14

##           Effective dimension: 2859

## [Scan]    9/51 | threshold = 0.16

##           Effective dimension: 2859

## [Scan]   10/51 | threshold = 0.18

##           Effective dimension: 2859

## [Scan]   11/51 | threshold = 0.2

##           Effective dimension: 2859

## [Scan]   12/51 | threshold = 0.22

##           Effective dimension: 2859

## [Scan]   13/51 | threshold = 0.24

##           Effective dimension: 2859

## [Scan]   14/51 | threshold = 0.26

##           Effective dimension: 2859

```

```
## [Scan] 15/51 | threshold = 0.28
##           Effective dimension: 2859
## [Scan] 16/51 | threshold = 0.3
##           Effective dimension: 2859
## [Scan] 17/51 | threshold = 0.32
##           Effective dimension: 2859
## [Scan] 18/51 | threshold = 0.34
##           Effective dimension: 2859
## [Scan] 19/51 | threshold = 0.36
##           Effective dimension: 2859
## [Scan] 20/51 | threshold = 0.38
##           Effective dimension: 2859
## [Scan] 21/51 | threshold = 0.4
##           Effective dimension: 2859
## [Scan] 22/51 | threshold = 0.42
##           Effective dimension: 2859
## [Scan] 23/51 | threshold = 0.44
##           Effective dimension: 2859
## [Scan] 24/51 | threshold = 0.46
##           Effective dimension: 2859
## [Scan] 25/51 | threshold = 0.48
```

```

##           Effective dimension: 2859

## [Scan]  26/51 | threshold = 0.5

##           Effective dimension: 2859

## [Scan]  27/51 | threshold = 0.52

##           Effective dimension: 2859

## [Scan]  28/51 | threshold = 0.54

##           Effective dimension: 2859

## [Scan]  29/51 | threshold = 0.56

##           Effective dimension: 2859

## [Scan]  30/51 | threshold = 0.58

##           Effective dimension: 2859

## [Scan]  31/51 | threshold = 0.6

##           Effective dimension: 2859

## [Scan]  32/51 | threshold = 0.62

##           Effective dimension: 2859

## [Scan]  33/51 | threshold = 0.64

##           Effective dimension: 2857

## [Scan]  34/51 | threshold = 0.66

##           Effective dimension: 2855

## [Scan]  35/51 | threshold = 0.68

##           Effective dimension: 2842

```

```
## [Scan] 36/51 | threshold = 0.7
##           Effective dimension: 2800
## [Scan] 37/51 | threshold = 0.72
##           Effective dimension: 2705
## [Scan] 38/51 | threshold = 0.74
##           Effective dimension: 2546
## [Scan] 39/51 | threshold = 0.76
##           Effective dimension: 2333
## [Scan] 40/51 | threshold = 0.78
##           Effective dimension: 2049
## [Scan] 41/51 | threshold = 0.8
##           Effective dimension: 1735
## [Scan] 42/51 | threshold = 0.82
##           Effective dimension: 1396
## [Scan] 43/51 | threshold = 0.84
##           Effective dimension: 1082
## [Scan] 44/51 | threshold = 0.86
##           Effective dimension: 825
## [Scan] 45/51 | threshold = 0.88
##           Effective dimension: 600
## [Scan] 46/51 | threshold = 0.9
```

```
##           Effective dimension: 454
```

```
## [Scan] 47/51 | threshold = 0.92
```

```
##           Effective dimension: 324
```

```
## [Scan] 48/51 | threshold = 0.94
```

```
##           Effective dimension: 215
```

```
## [Scan] 49/51 | threshold = 0.96
```

```
##           Effective dimension: 157
```

```
## [Scan] 50/51 | threshold = 0.98
```

```
##           Effective dimension: 79
```

```
## [Scan] 51/51 | threshold = 1
```

```
##           Effective dimension: 2
```

```
##           Too small. Stop scanning.
```

```
# The chosen threshold to feed downstream into build_graph_from_mat(r.threshold = ...)
out$chosen_threshold
```

```
## [1] 0.82
```

```
# ---- RMT threshold scan #2: plain `cor()` backend for comparison ----
# Same parameters as `out` above, but with `method = "cor"` (base stats::cor /
# psych::corr.test) instead of WGCNA. Useful as a sanity check - the chosen
# threshold should be very close across the two backends
out2 <- ggNetView_RMT(
  mat           = otu_rare_relative,
  transfrom.method = "none",
  method        = "cor",           # <-- only difference vs. `out`
  cor.method     = "pearson",
  unfold.method  = "gaussian",
  bandwidth      = "nrd0",
  nr.fit.points  = 20,
```

```
discard.outliers = TRUE,
discard.zeros    = TRUE,
min.mat.dim      = 40,
max.ev.spacing   = 3,
save_plots       = FALSE,
out_dir          = "RMT_plots",
verbose          = TRUE,
seed             = 1115
)

## [Info] Matrix dimension: 2859 x 2859

## [Info] #non-zeros: 8173761 | Sparseness: 0.0000

## [Scan] 1/51 | threshold = 0

## Effective dimension: 2859

## [Scan] 2/51 | threshold = 0.02

## Effective dimension: 2859

## [Scan] 3/51 | threshold = 0.04

## Effective dimension: 2859

## [Scan] 4/51 | threshold = 0.06

## Effective dimension: 2859

## [Scan] 5/51 | threshold = 0.08

## Effective dimension: 2859

## [Scan] 6/51 | threshold = 0.1

## Effective dimension: 2859

## [Scan] 7/51 | threshold = 0.12

## Effective dimension: 2859
```

```

## [Scan] 8/51 | threshold = 0.14

## Effective dimension: 2859

## [Scan] 9/51 | threshold = 0.16

## Effective dimension: 2859

## [Scan] 10/51 | threshold = 0.18

## Effective dimension: 2859

## [Scan] 11/51 | threshold = 0.2

## Effective dimension: 2859

## [Scan] 12/51 | threshold = 0.22

## Effective dimension: 2859

## [Scan] 13/51 | threshold = 0.24

## Effective dimension: 2859

## [Scan] 14/51 | threshold = 0.26

## Effective dimension: 2859

## [Scan] 15/51 | threshold = 0.28

## Effective dimension: 2859

## [Scan] 16/51 | threshold = 0.3

## Effective dimension: 2859

## [Scan] 17/51 | threshold = 0.32

## Effective dimension: 2859

## [Scan] 18/51 | threshold = 0.34

```



```
##           Effective dimension: 2859

## [Scan]  19/51 | threshold = 0.36

##           Effective dimension: 2859

## [Scan]  20/51 | threshold = 0.38

##           Effective dimension: 2859

## [Scan]  21/51 | threshold = 0.4

##           Effective dimension: 2859

## [Scan]  22/51 | threshold = 0.42

##           Effective dimension: 2859

## [Scan]  23/51 | threshold = 0.44

##           Effective dimension: 2859

## [Scan]  24/51 | threshold = 0.46

##           Effective dimension: 2859

## [Scan]  25/51 | threshold = 0.48

##           Effective dimension: 2859

## [Scan]  26/51 | threshold = 0.5

##           Effective dimension: 2859

## [Scan]  27/51 | threshold = 0.52

##           Effective dimension: 2859

## [Scan]  28/51 | threshold = 0.54

##           Effective dimension: 2859
```

```

## [Scan] 29/51 | threshold = 0.56

## Effective dimension: 2859

## [Scan] 30/51 | threshold = 0.58

## Effective dimension: 2859

## [Scan] 31/51 | threshold = 0.6

## Effective dimension: 2859

## [Scan] 32/51 | threshold = 0.62

## Effective dimension: 2859

## [Scan] 33/51 | threshold = 0.64

## Effective dimension: 2857

## [Scan] 34/51 | threshold = 0.66

## Effective dimension: 2855

## [Scan] 35/51 | threshold = 0.68

## Effective dimension: 2842

## [Scan] 36/51 | threshold = 0.7

## Effective dimension: 2800

## [Scan] 37/51 | threshold = 0.72

## Effective dimension: 2705

## [Scan] 38/51 | threshold = 0.74

## Effective dimension: 2546

## [Scan] 39/51 | threshold = 0.76

```

```
##           Effective dimension: 2333

## [Scan]  40/51 | threshold = 0.78

##           Effective dimension: 2049

## [Scan]  41/51 | threshold = 0.8

##           Effective dimension: 1735

## [Scan]  42/51 | threshold = 0.82

##           Effective dimension: 1396

## [Scan]  43/51 | threshold = 0.84

##           Effective dimension: 1082

## [Scan]  44/51 | threshold = 0.86

##           Effective dimension: 825

## [Scan]  45/51 | threshold = 0.88

##           Effective dimension: 600

## [Scan]  46/51 | threshold = 0.9

##           Effective dimension: 454

## [Scan]  47/51 | threshold = 0.92

##           Effective dimension: 324

## [Scan]  48/51 | threshold = 0.94

##           Effective dimension: 215

## [Scan]  49/51 | threshold = 0.96

##           Effective dimension: 157
```

```
## [Scan] 50/51 | threshold = 0.98

##           Effective dimension: 79

## [Scan] 51/51 | threshold = 1

##           Effective dimension: 10

##           Too small. Stop scanning.

# Compare with out$chosen_threshold - they should be in the same neighborhood.
out2$chosen_threshold

## [1] 0.82
```

3.4 Build graph from matrix based on RMT-threshold

Build graph object

```
# ---- Construct the tidygraph network using the RMT-selected r threshold ----
# Replaces the usual subjective r-cutoff (e.g. 0.6-0.85) with the data-driven
# value `out$chosen_threshold` returned above.
graph_obj <- build_graph_from_mat(
  mat                = otu_rare_relative,      # variable x sample matrix
  transform.method   = "none",                 # match the transform used for RMT scan
  r.threshold        = out$chosen_threshold,    # RMT-derived |r| cutoff (objective)
  p.threshold        = 0.05,                   # significance cutoff on adjusted p-value.
  method            = "WGCNA",                 # correlation backend (must match RMT scan)
  cor.method         = "pearson",               # correlation type
  proc              = "bonferroni",             # multiple-testing correction for p-value.
  module.method      = "Fast_greedy",           # community detection algorithm
  node_annotation    = tax_tab,                 # taxonomy joined onto each node (by name)
  top_modules        = 15,                      # keep top-15 modules; smaller ones -> "0"
  seed              = 1115                     # reproducibility
)

# Print to inspect node/edge counts and the columns now available on nodes
# (modularity, modularity2, modularity3, Modularity, Degree, Strength + taxonomy)
graph_obj
```

3.4. BUILD GRAPH FROM MATRIX BASED ON RMT-THRESHOLD 45

```
## # A tbl_graph: 213 nodes and 844 edges
## #
## # An undirected simple graph with 29 components
## #
## # Node Data: 213 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl> <chr>
## 1 ASV_649 5          5          5          5          27        26.5 Bacter~
## 2 ASV_705 5          5          5          5          27        26.5 Bacter~
## 3 ASV_12~ 5          5          5          5          27        26.5 Bacter~
## 4 ASV_13~ 5          5          5          5          27        26.5 Bacter~
## 5 ASV_14~ 5          5          5          5          27        26.5 Bacter~
## 6 ASV_14~ 5          5          5          5          27        26.5 Bacter~
## 7 ASV_24~ 5          5          5          5          27        26.5 Bacter~
## 8 ASV_25~ 5          5          5          5          27        26.4 Bacter~
## 9 ASV_28~ 5          5          5          5          27        26.5 Bacter~
## 10 ASV_28~ 5          5          5          5          27        26.5 Bacter~
## # i 203 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 844 x 5
##   from      to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1 194    195 0.959      0.959 Positive
## 2 185    208 0.954      0.954 Positive
## 3 185    213 0.957      0.957 Positive
## # i 841 more rows
```


Chapter 4

Get network information

Once the graph object is constructed, basic network information can be retrieved, and subnetworks can be extracted.

Load R Package

```
library(tidyverse)
library(ggNetView)
```

Example data

```
# Access built-in example datasets in ggNetView

# Relative abundance table of rarefied ASVs or OTUs
data("otu_rare_relative")
dim(otu_rare_relative)
```

```
## [1] 2859  18
```

```
otu_rare_relative[1:5, 1:5]
```

```
##           K01           K02           K03           K04           K05
## ASV_1 0.03306667 0.05453333 0.02013333 0.03613333 0.02686667
## ASV_2 0.05750000 0.03393333 0.06046667 0.05810000 0.07320000
## ASV_3 0.01733333 0.01296667 0.02290000 0.02336667 0.03106667
## ASV_4 0.04266667 0.01093333 0.01416667 0.01933333 0.03346667
## ASV_6 0.02646667 0.01856667 0.02110000 0.02353333 0.03806667
```

```
# Taxonomic annotation table of ASVs or OTUs
data("tax_tab")
dim(tax_tab)
```

```
## [1] 2859      8
```

```
tax_tab[1:5, 1:5]
```

```
## # A tibble: 5 x 5
##   OTUID Kingdom Phylum      Class      Order
##   <chr>  <chr>    <chr>    <chr>    <chr>
## 1 ASV_2  Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 2 ASV_3  Bacteria Verrucomicrobia Spartobacteria Unassigned
## 3 ASV_31 Bacteria Actinobacteria Actinobacteria Actinomycetales
## 4 ASV_27 Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 5 ASV_9  Bacteria Unassigned      Unassigned  Unassigned
```

Build graph object

```
graph_obj <- build_graph_from_mat(
  mat = otu_rare_relative,
  transform.method = "none", # based your input data
  r.threshold = 0.7,
  p.threshold = 0.05,
  method = "WGCNA",
  cor.method = "pearson",
  proc = "bonferroni",
  module.method = "Fast_greedy",
  node_annotation = tax_tab,
  top_modules = 15,
  seed = 1115
)
```

```
graph_obj
```

```
## # A tbl_graph: 213 nodes and 844 edges
```

```
## #
```

```
## # An undirected simple graph with 29 components
```

```
## #
```

```
## # Node Data: 213 x 14 (active)
```

```
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <dbl>      <dbl>      <dbl>      <dbl>    <dbl>    <dbl>  <chr>
## 1 ASV_649 5          5          5          5        27      26.5 Bacter~
```



```
## 2 ASV_705 5 5 5 5 27 26.5 Bacter~
## 3 ASV_12~ 5 5 5 5 27 26.5 Bacter~
## 4 ASV_13~ 5 5 5 5 27 26.5 Bacter~
## 5 ASV_14~ 5 5 5 5 27 26.5 Bacter~
## 6 ASV_14~ 5 5 5 5 27 26.5 Bacter~
## 7 ASV_24~ 5 5 5 5 27 26.5 Bacter~
## 8 ASV_25~ 5 5 5 5 27 26.4 Bacter~
## 9 ASV_28~ 5 5 5 5 27 26.5 Bacter~
## 10 ASV_28~ 5 5 5 5 27 26.5 Bacter~
## # i 203 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 844 x 5
## from to weight correlation corr_direction
## <int> <int> <dbl> <dbl> <chr>
## 1 194 195 0.959 0.959 Positive
## 2 185 208 0.954 0.954 Positive
## 3 185 213 0.957 0.957 Positive
## # i 841 more rows
```

4.1 Full-network information

```
graph_obj_info <- get_info_from_graph(graph_obj = graph_obj)
```

```
# return a list
class(graph_obj_info)
```

```
## [1] "list"
```

```
names(graph_obj_info)
```

```
## [1] "node_info" "edge_info"
```

```
graph_obj_info$node_info
```

```
## # A tibble: 213 x 11
## name Modularity Degree Strength Kingdom Phylum Class Order Family Genus
## <chr> <ord> <dbl> <dbl> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 ASV_649 5 27 26.5 Bacteria Proteo~ Alph~ Rhiz~ Unass~ Unas~
## 2 ASV_705 5 27 26.5 Bacteria Actino~ Acti~ Acti~ Cellu~ Cell~
```

```
## 3 ASV_1234 5 27 26.5 Bacteria Gemmat~ Gemm~ Gemm~ Gemma~ Gemm~
## 4 ASV_1373 5 27 26.5 Bacteria Actino~ Acti~ Soli~ Solir~ Soli~
## 5 ASV_1479 5 27 26.5 Bacteria Verruc~ Spar~ Unas~ Unass~ Spar~
## 6 ASV_1481 5 27 26.5 Bacteria Proteo~ Gamm~ Unas~ Unass~ Unas~
## 7 ASV_2453 5 27 26.5 Bacteria Bacter~ Sphi~ Sphi~ Chiti~ Unas~
## 8 ASV_2562 5 27 26.4 Bacteria Bacter~ Sphi~ Sphi~ Chiti~ Unas~
## 9 ASV_2848 5 27 26.5 Bacteria Planct~ Plan~ Plan~ Planc~ Unas~
## 10 ASV_2844 5 27 26.5 Bacteria Proteo~ Unas~ Unas~ Unass~ Unas~
## # i 203 more rows
## # i 1 more variable: Species <chr>
```

```
graph_obj_info$edge_info
```

```
## # A tibble: 844 x 5
##   from to weight correlation corr_direction
##   <chr> <chr> <dbl> <dbl> <chr>
## 1 ASV_6 ASV_39 0.959 0.959 Positive
## 2 ASV_20 ASV_1852 0.954 0.954 Positive
## 3 ASV_20 ASV_2911 0.957 0.957 Positive
## 4 ASV_172 ASV_43 0.969 0.969 Positive
## 5 ASV_614 ASV_65 0.943 0.943 Positive
## 6 ASV_69 ASV_173 0.995 0.995 Positive
## 7 ASV_69 ASV_206 0.994 0.994 Positive
## 8 ASV_69 ASV_744 0.976 0.976 Positive
## 9 ASV_945 ASV_69 0.985 0.985 Positive
## 10 ASV_69 ASV_1418 0.975 0.975 Positive
## # i 834 more rows
```

4.2 Sub-network (modularity) information

```
subgraph_info <- get_subgraph(graph_obj = graph_obj,
                               select_module = NULL)
```

```
##   Module Number
## 1      5      31
## 2      2      24
## 3      8      22
## 4     21      15
## 5      1      13
## 6      3      11
## 7      4      11
## 8      6      11
```

```
## 9      25      8
## 10     7      6
## 11     12     5
## 12     22     5
## 13     10     4
## 14     11     4
## 15     15     4
## 16 Others    39
```

```
# return a list
class(subgraph_info)
```

```
## [1] "list"
```

```
names(subgraph_info)
```

```
## [1] "sub_graph_all"      "stat_module"        "sub_graph_select"
```

```
subgraph_info$stat_module
```

```
##      Module Number
## 1         5      31
## 2         2      24
## 3         8      22
## 4        21      15
## 5         1      13
## 6         3      11
## 7         4      11
## 8         6      11
## 9        25       8
## 10        7       6
## 11       12       5
## 12       22       5
## 13       10       4
## 14       11       4
## 15       15       4
## 16 Others    39
```

```
subgraph_info$sub_graph_all
```

```
## $`5`
## # A tbl_graph: 31 nodes and 330 edges
```

```
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 31 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_649 5          5          5          5          27      26.5 Bacter~
## 2 ASV_705 5          5          5          5          27      26.5 Bacter~
## 3 ASV_12~ 5          5          5          5          27      26.5 Bacter~
## 4 ASV_13~ 5          5          5          5          27      26.5 Bacter~
## 5 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 6 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 7 ASV_24~ 5          5          5          5          27      26.5 Bacter~
## 8 ASV_25~ 5          5          5          5          27      26.4 Bacter~
## 9 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## 10 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## # i 21 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 330 x 5
##   from      to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.996      0.996 Positive
## 2      1      3 0.997      0.997 Positive
## 3      2      3 0.994      0.994 Positive
## # i 327 more rows
##
## $`2`
## # A tbl_graph: 24 nodes and 121 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 24 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_671 2          2          2          2          18      17.5 Bacter~
## 2 ASV_18~ 2          2          2          2          17      16.6 Bacter~
## 3 ASV_29~ 2          2          2          2          17      16.6 Bacter~
## 4 ASV_568 2          2          2          2          16      15.6 Bacter~
## 5 ASV_10~ 2          2          2          2          16      15.6 Bacter~
## 6 ASV_11~ 2          2          2          2          16      15.6 Bacter~
## 7 ASV_24~ 2          2          2          2          15      14.5 Bacter~
## 8 ASV_28~ 2          2          2          2          14      13.5 Bacter~
## 9 ASV_15~ 2          2          2          2          13      12.6 Bacter~
## 10 ASV_24~ 2          2          2          2          13      12.5 Bacter~
```

```
## # i 14 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 121 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.991        0.991 Positive
## 2     1     3  0.992        0.992 Positive
## 3     2     3  0.996        0.996 Positive
## # i 118 more rows
##
## $`8`
## # A tbl_graph: 22 nodes and 131 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 22 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_802 8          8          8          8          18      17.5 Bacter~
## 2 ASV_723 8          8          8          8          18      17.5 Bacter~
## 3 ASV_11~ 8          8          8          8          18      17.6 Bacter~
## 4 ASV_29~ 8          8          8          8          18      17.6 Bacter~
## 5 ASV_927 8          8          8          8          16      15.6 Bacter~
## 6 ASV_13~ 8          8          8          8          16      15.5 Bacter~
## 7 ASV_29~ 8          8          8          8          16      15.5 Bacter~
## 8 ASV_10~ 8          8          8          8          15      14.4 Bacter~
## 9 ASV_13~ 8          8          8          8          15      14.6 Bacter~
## 10 ASV_14~ 8          8          8          8          15      14.7 Bacter~
## # i 12 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 131 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.990        0.990 Positive
## 2     1     3  0.993        0.993 Positive
## 3     2     3  0.997        0.997 Positive
## # i 128 more rows
##
## $`21`
## # A tbl_graph: 15 nodes and 42 edges
## #
## # An undirected simple graph with 1 component
```

```
## #
## # Node Data: 15 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_25~ 21          21          21          21          13      12.5 Bacter~
## 2 ASV_27~ 21          21          21          21          12      11.6 Bacter~
## 3 ASV_879 21          21          21          21          8       7.67 Bacter~
## 4 ASV_10~ 21          21          21          21          6       5.76 Bacter~
## 5 ASV_16~ 21          21          21          21          6       5.70 Bacter~
## 6 ASV_511 21          21          21          21          5       4.80 Bacter~
## 7 ASV_578 21          21          21          21          5       4.76 Bacter~
## 8 ASV_917 21          21          21          21          5       4.80 Bacter~
## 9 ASV_18~ 21          21          21          21          5       4.81 Bacter~
## 10 ASV_70 21          21          21          21          4       3.82 Bacter~
## 11 ASV_13~ 21          21          21          21          4       3.86 Bacter~
## 12 ASV_839 21          21          21          21          3       2.85 Bacter~
## 13 ASV_20~ 21          21          21          21          3       2.86 Bacter~
## 14 ASV_21~ 21          21          21          21          3       2.86 Bacter~
## 15 ASV_88 21          21          21          21          2       1.90 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 42 x 5
##   from      to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.986        0.986 Positive
## 2      1      3 0.973        0.973 Positive
## 3      2      3 0.976        0.976 Positive
## # i 39 more rows
## #
## $`1`
## # A tbl_graph: 13 nodes and 43 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 13 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_19~ 1          1          1          1          9      8.86 Bacter~
## 2 ASV_15~ 1          1          1          1          9      8.86 Bacter~
## 3 ASV_24~ 1          1          1          1          9      8.87 Bacter~
## 4 ASV_20~ 1          1          1          1          9      8.79 Bacter~
## 5 ASV_20~ 1          1          1          1          9      8.87 Bacter~
## 6 ASV_26~ 1          1          1          1          9      8.87 Unassi~
## 7 ASV_968 1          1          1          1          8      7.88 Bacter~
## 8 ASV_27~ 1          1          1          1          8      7.73 Bacter~
```

```

## 9 ASV_161 1 1 1 1 7 6.80 Bacter~
## 10 ASV_25~ 1 1 1 1 5 4.72 Bacter~
## 11 ASV_534 1 1 1 1 2 1.90 Bacter~
## 12 ASV_26~ 1 1 1 1 1 0.944 Bacter~
## 13 ASV_29~ 1 1 1 1 1 0.963 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 43 x 5
## from to weight correlation corr_direction
## <int> <int> <dbl> <dbl> <chr>
## 1 1 2 0.998 0.998 Positive
## 2 1 3 0.999 0.999 Positive
## 3 2 3 1.000 1.000 Positive
## # i 40 more rows
##
## $`3`
## # A tbl_graph: 11 nodes and 38 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
## name modularity modularity2 modularity3 Modularity Degree Strength Kingdom
## <chr> <fct> <ord> <chr> <ord> <dbl> <dbl> <chr>
## 1 ASV_945 3 3 3 3 9 8.71 Bacter~
## 2 ASV_19~ 3 3 3 3 9 8.72 Bacter~
## 3 ASV_69 3 3 3 3 8 7.85 Bacter~
## 4 ASV_173 3 3 3 3 8 7.87 Bacter~
## 5 ASV_206 3 3 3 3 8 7.83 Bacter~
## 6 ASV_744 3 3 3 3 8 7.71 Bacter~
## 7 ASV_14~ 3 3 3 3 8 7.71 Bacter~
## 8 ASV_23~ 3 3 3 3 8 7.67 Bacter~
## 9 ASV_22~ 3 3 3 3 8 7.70 Bacter~
## 10 ASV_218 3 3 3 3 1 0.955 Bacter~
## 11 ASV_14~ 3 3 3 3 1 0.944 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 38 x 5
## from to weight correlation corr_direction
## <int> <int> <dbl> <dbl> <chr>
## 1 1 2 0.982 0.982 Positive
## 2 1 3 0.985 0.985 Positive
## 3 2 3 0.986 0.986 Positive
## # i 35 more rows
##

```

```
## $`4`
## # A tbl_graph: 11 nodes and 33 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_459 4          4          4          4          8      7.87 Bacter~
## 2 ASV_533 4          4          4          4          8      7.85 Bacter~
## 3 ASV_10~ 4          4          4          4          8      7.87 Bacter~
## 4 ASV_13~ 4          4          4          4          8      7.81 Bacter~
## 5 ASV_23~ 4          4          4          4          8      7.87 Bacter~
## 6 ASV_24~ 4          4          4          4          7      6.88 Bacter~
## 7 ASV_22~ 4          4          4          4          7      6.81 Bacter~
## 8 ASV_29~ 4          4          4          4          6      5.77 Bacter~
## 9 ASV_29~ 4          4          4          4          4      3.79 Bacter~
## 10 ASV_746 4          4          4          4          1      0.945 Bacter~
## 11 ASV_19~ 4          4          4          4          1      0.953 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 33 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.997      0.997 Positive
## 2      1      3 1.000      1.000 Positive
## 3      2      3 0.998      0.998 Positive
## # i 30 more rows
##
## $`6`
## # A tbl_graph: 11 nodes and 24 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_116 6          6          6          6          8      7.67 Bacter~
## 2 ASV_661 6          6          6          6          8      7.71 Bacter~
## 3 ASV_25~ 6          6          6          6          6      5.79 Bacter~
## 4 ASV_975 6          6          6          6          5      4.78 Bacter~
## 5 ASV_10~ 6          6          6          6          5      4.77 Bacter~
## 6 ASV_20~ 6          6          6          6          5      4.79 Bacter~
## 7 ASV_27~ 6          6          6          6          4      3.84 Bacter~
## 8 ASV_94  6          6          6          6          3      2.84 Bacter~
```



```

## 9 ASV_20~ 6          6          6          6          2      1.92 Bacter~
## 10 ASV_442 6          6          6          6          1      0.946 Bacter~
## 11 ASV_408 6          6          6          6          1      0.959 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 24 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2  0.974      0.974 Positive
## 2      1      3  0.966      0.966 Positive
## 3      2      3  0.979      0.979 Positive
## # i 21 more rows
##
## $`25`
## # A tbl_graph: 8 nodes and 22 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 8 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_1505 25          25          25          25          7        6.66 Bacter~
## 2 ASV_415  25          25          25          25          6        5.78 Bacter~
## 3 ASV_517  25          25          25          25          6        5.84 Bacter~
## 4 ASV_842  25          25          25          25          6        5.80 Bacter~
## 5 ASV_1719 25          25          25          25          6        5.85 Bacter~
## 6 ASV_2286 25          25          25          25          6        5.81 Bacter~
## 7 ASV_2637 25          25          25          25          5        4.78 Bacter~
## 8 ASV_1474 25          25          25          25          2        1.90 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 22 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2  0.947      0.947 Positive
## 2      1      3  0.951      0.951 Positive
## 3      2      3  0.970      0.970 Positive
## # i 19 more rows
##
## $`7`
## # A tbl_graph: 6 nodes and 5 edges
## #
## # An unrooted tree
## #

```

```
## # Node Data: 6 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_1896 7          7          7          7          3      2.87 Bacter~
## 2 ASV_454  7          7          7          7          2      1.89 Bacter~
## 3 ASV_1090 7          7          7          7          2      1.89 Bacter~
## 4 ASV_231  7          7          7          7          1      0.968 Bacter~
## 5 ASV_1800 7          7          7          7          1      0.943 Bacter~
## 6 ASV_2038 7          7          7          7          1      0.956 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 5 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2 0.945      0.945 Positive
## 2     2     3 0.944      0.944 Positive
## 3     1     4 0.968      0.968 Positive
## # i 2 more rows
##
## $`12`
## # A tbl_graph: 5 nodes and 5 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 5 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_1148 12          12          12          12          4      3.85 Bacter~
## 2 ASV_1104 12          12          12          12          2      1.92 Bacter~
## 3 ASV_1854 12          12          12          12          2      1.95 Bacter~
## 4 ASV_115  12          12          12          12          1      0.955 Bacter~
## 5 ASV_2484 12          12          12          12          1      0.948 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 5 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2 0.959      0.959 Positive
## 2     1     3 0.990      0.990 Positive
## 3     2     3 0.957      0.957 Positive
## # i 2 more rows
##
## $`22`
## # A tbl_graph: 5 nodes and 7 edges
```

```
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 5 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_1931 22          22          22          22          4        3.88 Bacter~
## 2 ASV_1320 22          22          22          22          3        2.89 Bacter~
## 3 ASV_1906 22          22          22          22          3        2.91 Bacter~
## 4 ASV_2704 22          22          22          22          3        2.91 Bacter~
## 5 ASV_1495 22          22          22          22          1        0.970 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 7 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.971      0.971 Positive
## 2     1     3  0.969      0.969 Positive
## 3     2     3  0.959      0.959 Positive
## # i 4 more rows
## #
## $`10`
## # A tbl_graph: 4 nodes and 3 edges
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_2059 10          10          10          10          2        1.89 Bacter~
## 2 ASV_1965 10          10          10          10          2        1.91 Bacter~
## 3 ASV_544  10          10          10          10          1        0.964 Bacter~
## 4 ASV_1607 10          10          10          10          1        0.946 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.943      0.943 Positive
## 2     2     3  0.964      0.964 Positive
## 3     1     4  0.946      0.946 Positive
## #
## $`11`
## # A tbl_graph: 4 nodes and 3 edges
```

```
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_614  11          11          11          11          2      1.91 Bacter~
## 2 ASV_2916 11          11          11          11          2      1.93 Bacter~
## 3 ASV_65    11          11          11          11          1      0.943 Archaea
## 4 ASV_2168 11          11          11          11          1      0.965 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.966        0.966 Positive
## 2      1      3 0.943        0.943 Positive
## 3      2      4 0.965        0.965 Positive
##
## $`15`
## # A tbl_graph: 4 nodes and 3 edges
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_172  15          15          15          15          2      1.92 Bacter~
## 2 ASV_246  15          15          15          15          2      1.91 Bacter~
## 3 ASV_43    15          15          15          15          1      0.969 Bacter~
## 4 ASV_2211 15          15          15          15          1      0.955 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.952        0.952 Positive
## 2      1      3 0.969        0.969 Positive
## 3      2      4 0.955        0.955 Positive
##
## $0others
## # A tbl_graph: 39 nodes and 34 edges
## #
## # An undirected simple graph with 14 components
```

```
## #
## # Node Data: 39 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>     <dbl> <chr>
## 1 ASV_316 28          Others    Others    Others      3        2.89 Bacter~
## 2 ASV_610 28          Others    Others    Others      3        2.94 Bacter~
## 3 ASV_11~ 29          Others    Others    Others      3        2.94 Bacter~
## 4 ASV_18~ 29          Others    Others    Others      3        2.95 Bacter~
## 5 ASV_11~ 27          Others    Others    Others      3        2.91 Bacter~
## 6 ASV_18~ 28          Others    Others    Others      3        2.91 Bacter~
## 7 ASV_20~ 29          Others    Others    Others      3        2.89 Bacter~
## 8 ASV_24~ 27          Others    Others    Others      3        2.89 Bacter~
## 9 ASV_27~ 29          Others    Others    Others      3        2.95 Bacter~
## 10 ASV_28~ 28          Others    Others    Others      3        2.90 Bacter~
## # i 29 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 34 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.971      0.971 Positive
## 2     3     4  0.987      0.987 Positive
## 3     1     6  0.958      0.958 Positive
## # i 31 more rows
```

```
subgraph_info$sub_graph_select
```

```
## NULL
```

4.3 Sub-network (sample) information

Chapter 5

Extract subgraph

Load R Package

```
library(tidyverse)
library(ggNetView)
```

Example data

```
# Access built-in example datasets in ggNetView

# Relative abundance table of rarefied ASVs or OTUs
data("otu_rare_relative")
dim(otu_rare_relative)
```

```
## [1] 2859  18
```

```
otu_rare_relative[1:5, 1:5]
```

```
##           K01           K02           K03           K04           K05
## ASV_1 0.03306667 0.05453333 0.02013333 0.03613333 0.02686667
## ASV_2 0.05750000 0.03393333 0.06046667 0.05810000 0.07320000
## ASV_3 0.01733333 0.01296667 0.02290000 0.02336667 0.03106667
## ASV_4 0.04266667 0.01093333 0.01416667 0.01933333 0.03346667
## ASV_6 0.02646667 0.01856667 0.02110000 0.02353333 0.03806667
```

```
# Taxonomic annotation table of ASVs or OTUs
data("tax_tab")
dim(tax_tab)
```

```
## [1] 2859      8
```

```
tax_tab[1:5, 1:5]
```

```
## # A tibble: 5 x 5
##   OTUID Kingdom Phylum      Class      Order
##   <chr>  <chr>    <chr>      <chr>      <chr>
## 1 ASV_2  Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 2 ASV_3  Bacteria Verrucomicrobia Spartobacteria Unassigned
## 3 ASV_31 Bacteria Actinobacteria Actinobacteria Actinomycetales
## 4 ASV_27 Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 5 ASV_9  Bacteria Unassigned      Unassigned  Unassigned
```

Build graph object

```
graph_obj <- build_graph_from_mat(
  mat = otu_rare_relative,
  transform.method = "none", # based your input data
  r.threshold = 0.7,
  p.threshold = 0.05,
  method = "WGCNA",
  cor.method = "pearson",
  proc = "bonferroni",
  module.method = "Fast_greedy",
  node_annotation = tax_tab,
  top_modules = 15,
  seed = 1115
)

graph_obj
```

```
## # A tbl_graph: 213 nodes and 844 edges
```

```
## #
```

```
## # An undirected simple graph with 29 components
```

```
## #
```

```
## # Node Data: 213 x 14 (active)
```

	name	modularity	modularity2	modularity3	Modularity	Degree	Strength	Kingdom
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
## 1	ASV_649	5	5	5	5	27	26.5	Bacter~
## 2	ASV_705	5	5	5	5	27	26.5	Bacter~
## 3	ASV_12~	5	5	5	5	27	26.5	Bacter~
## 4	ASV_13~	5	5	5	5	27	26.5	Bacter~
## 5	ASV_14~	5	5	5	5	27	26.5	Bacter~
## 6	ASV_14~	5	5	5	5	27	26.5	Bacter~


```
## 7 ASV_24~ 5          5          5          5          27      26.5 Bacter~
## 8 ASV_25~ 5          5          5          5          27      26.4 Bacter~
## 9 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## 10 ASV_28~ 5         5          5          5          27      26.5 Bacter~
## # i 203 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 844 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>         <dbl> <chr>
## 1    194   195  0.959         0.959 Positive
## 2    185   208  0.954         0.954 Positive
## 3    185   213  0.957         0.957 Positive
## # i 841 more rows
```

5.1 Extract subgraph by module

5.1.1 select subgraph by modularity

Select one modularity

```
module5_subgraph_info <- get_subgraph(graph_obj = graph_obj,
                                       select_module = "5")
```

```
##   Module Number
## 1         5    31
## 2         2    24
## 3         8    22
## 4        21    15
## 5         1    13
## 6         3    11
## 7         4    11
## 8         6    11
## 9        25     8
## 10        7     6
## 11       12     5
## 12       22     5
## 13       10     4
## 14       11     4
## 15       15     4
## 16 Others    39
```

```
class(module5_subgraph_info)
```

```
## [1] "list"
```

```
names(module5_subgraph_info)
```

```
## [1] "sub_graph_all"      "stat_module"        "sub_graph_select"
```

```
module5_subgraph_info$stat_module
```

```
##      Module Number
## 1         5      31
## 2         2      24
## 3         8      22
## 4        21      15
## 5         1      13
## 6         3      11
## 7         4      11
## 8         6      11
## 9        25       8
## 10        7       6
## 11       12       5
## 12       22       5
## 13       10       4
## 14       11       4
## 15       15       4
## 16 Others      39
```

```
module5_subgraph_info$sub_graph_all
```

```
## $`5`
```

```
## # A tbl_graph: 31 nodes and 330 edges
```

```
## #
```

```
## # An undirected simple graph with 1 component
```

```
## #
```

```
## # Node Data: 31 x 14 (active)
```

##	name	modularity	modularity2	modularity3	Modularity	Degree	Strength	Kingdom
##	<chr>	<fct>	<ord>	<chr>	<ord>	<dbl>	<dbl>	<chr>
##	1 ASV_649	5	5	5	5	27	26.5	Bacter~
##	2 ASV_705	5	5	5	5	27	26.5	Bacter~
##	3 ASV_12~	5	5	5	5	27	26.5	Bacter~
##	4 ASV_13~	5	5	5	5	27	26.5	Bacter~

```

## 5 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 6 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 7 ASV_24~ 5          5          5          5          27      26.5 Bacter~
## 8 ASV_25~ 5          5          5          5          27      26.4 Bacter~
## 9 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## 10 ASV_28~ 5         5          5          5          27      26.5 Bacter~
## # i 21 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 330 x 5
##   from   to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.996      0.996 Positive
## 2     1     3  0.997      0.997 Positive
## 3     2     3  0.994      0.994 Positive
## # i 327 more rows
##
## $`2`
## # A tbl_graph: 24 nodes and 121 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 24 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>   <dbl> <chr>
## 1 ASV_671 2          2          2          2          18     17.5 Bacter~
## 2 ASV_18~ 2          2          2          2          17     16.6 Bacter~
## 3 ASV_29~ 2          2          2          2          17     16.6 Bacter~
## 4 ASV_568 2          2          2          2          16     15.6 Bacter~
## 5 ASV_10~ 2          2          2          2          16     15.6 Bacter~
## 6 ASV_11~ 2          2          2          2          16     15.6 Bacter~
## 7 ASV_24~ 2          2          2          2          15     14.5 Bacter~
## 8 ASV_28~ 2          2          2          2          14     13.5 Bacter~
## 9 ASV_15~ 2          2          2          2          13     12.6 Bacter~
## 10 ASV_24~ 2          2          2          2          13     12.5 Bacter~
## # i 14 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 121 x 5
##   from   to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.991      0.991 Positive
## 2     1     3  0.992      0.992 Positive
## 3     2     3  0.996      0.996 Positive

```

```
## # i 118 more rows
##
## $`8`
## # A tbl_graph: 22 nodes and 131 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 22 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_802 8          8          8          8          18      17.5 Bacter~
## 2 ASV_723 8          8          8          8          18      17.5 Bacter~
## 3 ASV_11~ 8          8          8          8          18      17.6 Bacter~
## 4 ASV_29~ 8          8          8          8          18      17.6 Bacter~
## 5 ASV_927 8          8          8          8          16      15.6 Bacter~
## 6 ASV_13~ 8          8          8          8          16      15.5 Bacter~
## 7 ASV_29~ 8          8          8          8          16      15.5 Bacter~
## 8 ASV_10~ 8          8          8          8          15      14.4 Bacter~
## 9 ASV_13~ 8          8          8          8          15      14.6 Bacter~
## 10 ASV_14~ 8          8          8          8          15      14.7 Bacter~
## # i 12 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 131 x 5
##   from      to weight correlation corr_direction
##   <int> <int>  <dbl>      <dbl> <chr>
## 1      1      2 0.990      0.990 Positive
## 2      1      3 0.993      0.993 Positive
## 3      2      3 0.997      0.997 Positive
## # i 128 more rows
##
## $`21`
## # A tbl_graph: 15 nodes and 42 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 15 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_25~ 21          21          21          21          13      12.5 Bacter~
## 2 ASV_27~ 21          21          21          21          12      11.6 Bacter~
## 3 ASV_879 21          21          21          21          8       7.67 Bacter~
## 4 ASV_10~ 21          21          21          21          6       5.76 Bacter~
## 5 ASV_16~ 21          21          21          21          6       5.70 Bacter~
## 6 ASV_511 21          21          21          21          5       4.80 Bacter~
```

```

## 7 ASV_578 21          21          21          21          5      4.76 Bacter~
## 8 ASV_917 21          21          21          21          5      4.80 Bacter~
## 9 ASV_18~ 21          21          21          21          5      4.81 Bacter~
## 10 ASV_70 21          21          21          21          4      3.82 Bacter~
## 11 ASV_13~ 21          21          21          21          4      3.86 Bacter~
## 12 ASV_839 21          21          21          21          3      2.85 Bacter~
## 13 ASV_20~ 21          21          21          21          3      2.86 Bacter~
## 14 ASV_21~ 21          21          21          21          3      2.86 Bacter~
## 15 ASV_88 21          21          21          21          2      1.90 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 42 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>         <dbl> <chr>
## 1      1      2  0.986         0.986 Positive
## 2      1      3  0.973         0.973 Positive
## 3      2      3  0.976         0.976 Positive
## # i 39 more rows
##
## $`1`
## # A tbl_graph: 13 nodes and 43 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 13 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>       <ord>       <chr>       <ord>       <dbl>   <dbl> <chr>
## 1 ASV_19~ 1           1           1           1           9      8.86 Bacter~
## 2 ASV_15~ 1           1           1           1           9      8.86 Bacter~
## 3 ASV_24~ 1           1           1           1           9      8.87 Bacter~
## 4 ASV_20~ 1           1           1           1           9      8.79 Bacter~
## 5 ASV_20~ 1           1           1           1           9      8.87 Bacter~
## 6 ASV_26~ 1           1           1           1           9      8.87 Unassi~
## 7 ASV_968 1           1           1           1           8      7.88 Bacter~
## 8 ASV_27~ 1           1           1           1           8      7.73 Bacter~
## 9 ASV_161 1           1           1           1           7      6.80 Bacter~
## 10 ASV_25~ 1           1           1           1           5      4.72 Bacter~
## 11 ASV_534 1           1           1           1           2      1.90 Bacter~
## 12 ASV_26~ 1           1           1           1           1      0.944 Bacter~
## 13 ASV_29~ 1           1           1           1           1      0.963 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 43 x 5
##   from    to weight correlation corr_direction

```

```

##      <int> <int>      <dbl>          <dbl> <chr>
## 1      1      2 0.998      0.998 Positive
## 2      1      3 0.999      0.999 Positive
## 3      2      3 1.000      1.000 Positive
## # i 40 more rows
##
## $`3`
## # A tbl_graph: 11 nodes and 38 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_945 3          3          3          3          9      8.71 Bacter~
## 2 ASV_19~ 3          3          3          3          9      8.72 Bacter~
## 3 ASV_69 3          3          3          3          8      7.85 Bacter~
## 4 ASV_173 3          3          3          3          8      7.87 Bacter~
## 5 ASV_206 3          3          3          3          8      7.83 Bacter~
## 6 ASV_744 3          3          3          3          8      7.71 Bacter~
## 7 ASV_14~ 3          3          3          3          8      7.71 Bacter~
## 8 ASV_23~ 3          3          3          3          8      7.67 Bacter~
## 9 ASV_22~ 3          3          3          3          8      7.70 Bacter~
## 10 ASV_218 3          3          3          3          1      0.955 Bacter~
## 11 ASV_14~ 3          3          3          3          1      0.944 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 38 x 5
##   from      to weight correlation corr_direction
##   <int> <int>      <dbl>          <dbl> <chr>
## 1      1      2 0.982      0.982 Positive
## 2      1      3 0.985      0.985 Positive
## 3      2      3 0.986      0.986 Positive
## # i 35 more rows
##
## $`4`
## # A tbl_graph: 11 nodes and 33 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_459 4          4          4          4          8      7.87 Bacter~
## 2 ASV_533 4          4          4          4          8      7.85 Bacter~

```

```

## 3 ASV_10~ 4      4      4      4      8      7.87 Bacter~
## 4 ASV_13~ 4      4      4      4      8      7.81 Bacter~
## 5 ASV_23~ 4      4      4      4      8      7.87 Bacter~
## 6 ASV_24~ 4      4      4      4      7      6.88 Bacter~
## 7 ASV_22~ 4      4      4      4      7      6.81 Bacter~
## 8 ASV_29~ 4      4      4      4      6      5.77 Bacter~
## 9 ASV_29~ 4      4      4      4      4      3.79 Bacter~
## 10 ASV_746 4      4      4      4      1      0.945 Bacter~
## 11 ASV_19~ 4      4      4      4      1      0.953 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 33 x 5
##   from   to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.997      0.997 Positive
## 2     1     3  1.000      1.000 Positive
## 3     2     3  0.998      0.998 Positive
## # i 30 more rows
##
## $`6`
## # A tbl_graph: 11 nodes and 24 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_116 6          6          6          6          8      7.67 Bacter~
## 2 ASV_661 6          6          6          6          8      7.71 Bacter~
## 3 ASV_25~ 6          6          6          6          6      5.79 Bacter~
## 4 ASV_975 6          6          6          6          5      4.78 Bacter~
## 5 ASV_10~ 6          6          6          6          5      4.77 Bacter~
## 6 ASV_20~ 6          6          6          6          5      4.79 Bacter~
## 7 ASV_27~ 6          6          6          6          4      3.84 Bacter~
## 8 ASV_94  6          6          6          6          3      2.84 Bacter~
## 9 ASV_20~ 6          6          6          6          2      1.92 Bacter~
## 10 ASV_442 6          6          6          6          1      0.946 Bacter~
## 11 ASV_408 6          6          6          6          1      0.959 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 24 x 5
##   from   to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.974      0.974 Positive

```

```

## 2      1      3 0.966      0.966 Positive
## 3      2      3 0.979      0.979 Positive
## # i 21 more rows
##
## $`25`
## # A tbl_graph: 8 nodes and 22 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 8 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_1505 25          25          25          25          7          6.66 Bacter~
## 2 ASV_415  25          25          25          25          6          5.78 Bacter~
## 3 ASV_517  25          25          25          25          6          5.84 Bacter~
## 4 ASV_842  25          25          25          25          6          5.80 Bacter~
## 5 ASV_1719 25          25          25          25          6          5.85 Bacter~
## 6 ASV_2286 25          25          25          25          6          5.81 Bacter~
## 7 ASV_2637 25          25          25          25          5          4.78 Bacter~
## 8 ASV_1474 25          25          25          25          2          1.90 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 22 x 5
##   from      to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.947      0.947 Positive
## 2      1      3 0.951      0.951 Positive
## 3      2      3 0.970      0.970 Positive
## # i 19 more rows
##
## $`7`
## # A tbl_graph: 6 nodes and 5 edges
## #
## # An unrooted tree
## #
## # Node Data: 6 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_1896 7          7          7          7          3          2.87 Bacter~
## 2 ASV_454  7          7          7          7          2          1.89 Bacter~
## 3 ASV_1090 7          7          7          7          2          1.89 Bacter~
## 4 ASV_231  7          7          7          7          1          0.968 Bacter~
## 5 ASV_1800 7          7          7          7          1          0.943 Bacter~
## 6 ASV_2038 7          7          7          7          1          0.956 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,

```



```
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 5 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.945      0.945 Positive
## 2     2     3  0.944      0.944 Positive
## 3     1     4  0.968      0.968 Positive
## # i 2 more rows
##
## $`12`
## # A tbl_graph: 5 nodes and 5 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 5 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_1148 12          12          12          12          4        3.85 Bacter~
## 2 ASV_1104 12          12          12          12          2        1.92 Bacter~
## 3 ASV_1854 12          12          12          12          2        1.95 Bacter~
## 4 ASV_115  12          12          12          12          1        0.955 Bacter~
## 5 ASV_2484 12          12          12          12          1        0.948 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 5 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.959      0.959 Positive
## 2     1     3  0.990      0.990 Positive
## 3     2     3  0.957      0.957 Positive
## # i 2 more rows
##
## $`22`
## # A tbl_graph: 5 nodes and 7 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 5 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_1931 22          22          22          22          4        3.88 Bacter~
## 2 ASV_1320 22          22          22          22          3        2.89 Bacter~
## 3 ASV_1906 22          22          22          22          3        2.91 Bacter~
## 4 ASV_2704 22          22          22          22          3        2.91 Bacter~
```

```

## 5 ASV_1495 22          22          22          22          1    0.970 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 7 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.971      0.971 Positive
## 2     1     3  0.969      0.969 Positive
## 3     2     3  0.959      0.959 Positive
## # i 4 more rows
##
## $`10`
## # A tbl_graph: 4 nodes and 3 edges
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_2059 10          10          10          10          2      1.89 Bacter~
## 2 ASV_1965 10          10          10          10          2      1.91 Bacter~
## 3 ASV_544  10          10          10          10          1      0.964 Bacter~
## 4 ASV_1607 10          10          10          10          1      0.946 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.943      0.943 Positive
## 2     2     3  0.964      0.964 Positive
## 3     1     4  0.946      0.946 Positive
##
## $`11`
## # A tbl_graph: 4 nodes and 3 edges
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_614  11          11          11          11          2      1.91 Bacter~
## 2 ASV_2916 11          11          11          11          2      1.93 Bacter~
## 3 ASV_65    11          11          11          11          1      0.943 Archaea
## 4 ASV_2168 11          11          11          11          1      0.965 Bacter~

```

```

## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.966      0.966 Positive
## 2     1     3  0.943      0.943 Positive
## 3     2     4  0.965      0.965 Positive
##
## $`15`
## # A tbl_graph: 4 nodes and 3 edges
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_172   15          15          15          15          2          1.92 Bacter~
## 2 ASV_246   15          15          15          15          2          1.91 Bacter~
## 3 ASV_43    15          15          15          15          1          0.969 Bacter~
## 4 ASV_2211  15          15          15          15          1          0.955 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.952      0.952 Positive
## 2     1     3  0.969      0.969 Positive
## 3     2     4  0.955      0.955 Positive
##
## $Others
## # A tbl_graph: 39 nodes and 34 edges
## #
## # An undirected simple graph with 14 components
## #
## # Node Data: 39 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_316   28          Others    Others    Others    3          2.89 Bacter~
## 2 ASV_610   28          Others    Others    Others    3          2.94 Bacter~
## 3 ASV_11~   29          Others    Others    Others    3          2.94 Bacter~
## 4 ASV_18~   29          Others    Others    Others    3          2.95 Bacter~
## 5 ASV_11~   27          Others    Others    Others    3          2.91 Bacter~
## 6 ASV_18~   28          Others    Others    Others    3          2.91 Bacter~

```

```
## 7 ASV_20~ 29      Others      Others      Others      3      2.89 Bacter~
## 8 ASV_24~ 27      Others      Others      Others      3      2.89 Bacter~
## 9 ASV_27~ 29      Others      Others      Others      3      2.95 Bacter~
## 10 ASV_28~ 28     Others      Others      Others      3      2.90 Bacter~
## # i 29 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 34 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.971      0.971 Positive
## 2      3      4 0.987      0.987 Positive
## 3      1      6 0.958      0.958 Positive
## # i 31 more rows
```

```
module5_subgraph_info$sub_graph_select
```

```
## # A tbl_graph: 31 nodes and 330 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 31 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_649 5          5          5          5          27      26.5 Bacter~
## 2 ASV_705 5          5          5          5          27      26.5 Bacter~
## 3 ASV_12~ 5          5          5          5          27      26.5 Bacter~
## 4 ASV_13~ 5          5          5          5          27      26.5 Bacter~
## 5 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 6 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 7 ASV_24~ 5          5          5          5          27      26.5 Bacter~
## 8 ASV_25~ 5          5          5          5          27      26.4 Bacter~
## 9 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## 10 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## # i 21 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 330 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      29    30 0.949      0.949 Positive
## 2      21    29 0.952      0.952 Positive
## 3      25    29 0.965      0.965 Positive
```

```
## # i 327 more rows
```

Select multi-modularity

```
module5_2_8_subgraph_info <- get_subgraph(graph_obj = graph_obj,
                                           select_module = c("5","2","8"))
```

```
##      Module Number
## 1         5      31
## 2         2      24
## 3         8      22
## 4        21      15
## 5         1      13
## 6         3      11
## 7         4      11
## 8         6      11
## 9        25       8
## 10        7       6
## 11       12       5
## 12       22       5
## 13       10       4
## 14       11       4
## 15       15       4
## 16 Others      39
```

```
class(module5_2_8_subgraph_info)
```

```
## [1] "list"
```

```
names(module5_2_8_subgraph_info)
```

```
## [1] "sub_graph_all"      "stat_module"        "sub_graph_select"
```

```
module5_2_8_subgraph_info$stat_module
```

```
##      Module Number
## 1         5      31
## 2         2      24
## 3         8      22
## 4        21      15
## 5         1      13
## 6         3      11
```

```
## 7      4      11
## 8      6      11
## 9     25      8
## 10     7      6
## 11    12      5
## 12    22      5
## 13    10      4
## 14    11      4
## 15    15      4
## 16 Others    39
```

```
module5_2_8_subgraph_info$sub_graph_all
```

```
## $`5`
## # A tbl_graph: 31 nodes and 330 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 31 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_649 5          5          5          5          27      26.5 Bacter~
## 2 ASV_705 5          5          5          5          27      26.5 Bacter~
## 3 ASV_12~ 5          5          5          5          27      26.5 Bacter~
## 4 ASV_13~ 5          5          5          5          27      26.5 Bacter~
## 5 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 6 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 7 ASV_24~ 5          5          5          5          27      26.5 Bacter~
## 8 ASV_25~ 5          5          5          5          27      26.4 Bacter~
## 9 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## 10 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## # i 21 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 330 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.996      0.996 Positive
## 2      1      3 0.997      0.997 Positive
## 3      2      3 0.994      0.994 Positive
## # i 327 more rows
##
## $`2`
## # A tbl_graph: 24 nodes and 121 edges
```

```
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 24 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_671 2          2          2          2          18      17.5 Bacter~
## 2 ASV_18~ 2          2          2          2          17      16.6 Bacter~
## 3 ASV_29~ 2          2          2          2          17      16.6 Bacter~
## 4 ASV_568 2          2          2          2          16      15.6 Bacter~
## 5 ASV_10~ 2          2          2          2          16      15.6 Bacter~
## 6 ASV_11~ 2          2          2          2          16      15.6 Bacter~
## 7 ASV_24~ 2          2          2          2          15      14.5 Bacter~
## 8 ASV_28~ 2          2          2          2          14      13.5 Bacter~
## 9 ASV_15~ 2          2          2          2          13      12.6 Bacter~
## 10 ASV_24~ 2          2          2          2          13      12.5 Bacter~
## # i 14 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 121 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2 0.991      0.991 Positive
## 2     1     3 0.992      0.992 Positive
## 3     2     3 0.996      0.996 Positive
## # i 118 more rows
##
## $`8`
## # A tbl_graph: 22 nodes and 131 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 22 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_802 8          8          8          8          18      17.5 Bacter~
## 2 ASV_723 8          8          8          8          18      17.5 Bacter~
## 3 ASV_11~ 8          8          8          8          18      17.6 Bacter~
## 4 ASV_29~ 8          8          8          8          18      17.6 Bacter~
## 5 ASV_927 8          8          8          8          16      15.6 Bacter~
## 6 ASV_13~ 8          8          8          8          16      15.5 Bacter~
## 7 ASV_29~ 8          8          8          8          16      15.5 Bacter~
## 8 ASV_10~ 8          8          8          8          15      14.4 Bacter~
## 9 ASV_13~ 8          8          8          8          15      14.6 Bacter~
## 10 ASV_14~ 8          8          8          8          15      14.7 Bacter~
```

```

## # i 12 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 131 x 5
##   from      to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2  0.990        0.990 Positive
## 2      1      3  0.993        0.993 Positive
## 3      2      3  0.997        0.997 Positive
## # i 128 more rows
##
## $`21`
## # A tbl_graph: 15 nodes and 42 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 15 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_25~ 21          21          21          21          13      12.5 Bacter~
## 2 ASV_27~ 21          21          21          21          12      11.6 Bacter~
## 3 ASV_879 21          21          21          21          8       7.67 Bacter~
## 4 ASV_10~ 21          21          21          21          6       5.76 Bacter~
## 5 ASV_16~ 21          21          21          21          6       5.70 Bacter~
## 6 ASV_511 21          21          21          21          5       4.80 Bacter~
## 7 ASV_578 21          21          21          21          5       4.76 Bacter~
## 8 ASV_917 21          21          21          21          5       4.80 Bacter~
## 9 ASV_18~ 21          21          21          21          5       4.81 Bacter~
## 10 ASV_70 21          21          21          21          4       3.82 Bacter~
## 11 ASV_13~ 21          21          21          21          4       3.86 Bacter~
## 12 ASV_839 21          21          21          21          3       2.85 Bacter~
## 13 ASV_20~ 21          21          21          21          3       2.86 Bacter~
## 14 ASV_21~ 21          21          21          21          3       2.86 Bacter~
## 15 ASV_88 21          21          21          21          2       1.90 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 42 x 5
##   from      to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2  0.986        0.986 Positive
## 2      1      3  0.973        0.973 Positive
## 3      2      3  0.976        0.976 Positive
## # i 39 more rows
##

```



```
## $`1`
## # A tbl_graph: 13 nodes and 43 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 13 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_19~ 1          1          1          1          9      8.86 Bacter~
## 2 ASV_15~ 1          1          1          1          9      8.86 Bacter~
## 3 ASV_24~ 1          1          1          1          9      8.87 Bacter~
## 4 ASV_20~ 1          1          1          1          9      8.79 Bacter~
## 5 ASV_20~ 1          1          1          1          9      8.87 Bacter~
## 6 ASV_26~ 1          1          1          1          9      8.87 Unassi~
## 7 ASV_968 1          1          1          1          8      7.88 Bacter~
## 8 ASV_27~ 1          1          1          1          8      7.73 Bacter~
## 9 ASV_161 1          1          1          1          7      6.80 Bacter~
## 10 ASV_25~ 1          1          1          1          5      4.72 Bacter~
## 11 ASV_534 1          1          1          1          2      1.90 Bacter~
## 12 ASV_26~ 1          1          1          1          1      0.944 Bacter~
## 13 ASV_29~ 1          1          1          1          1      0.963 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 43 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.998      0.998 Positive
## 2     1     3  0.999      0.999 Positive
## 3     2     3  1.000      1.000 Positive
## # i 40 more rows
##
## $`3`
## # A tbl_graph: 11 nodes and 38 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_945 3          3          3          3          9      8.71 Bacter~
## 2 ASV_19~ 3          3          3          3          9      8.72 Bacter~
## 3 ASV_69 3          3          3          3          8      7.85 Bacter~
## 4 ASV_173 3          3          3          3          8      7.87 Bacter~
## 5 ASV_206 3          3          3          3          8      7.83 Bacter~
## 6 ASV_744 3          3          3          3          8      7.71 Bacter~
```

```

## 7 ASV_14~ 3      3      3      3      8      7.71 Bacter~
## 8 ASV_23~ 3      3      3      3      8      7.67 Bacter~
## 9 ASV_22~ 3      3      3      3      8      7.70 Bacter~
## 10 ASV_218 3      3      3      3      1      0.955 Bacter~
## 11 ASV_14~ 3      3      3      3      1      0.944 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 38 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.982      0.982 Positive
## 2      1      3 0.985      0.985 Positive
## 3      2      3 0.986      0.986 Positive
## # i 35 more rows
##
## $`4`
## # A tbl_graph: 11 nodes and 33 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_459 4          4          4          4          8      7.87 Bacter~
## 2 ASV_533 4          4          4          4          8      7.85 Bacter~
## 3 ASV_10~ 4          4          4          4          8      7.87 Bacter~
## 4 ASV_13~ 4          4          4          4          8      7.81 Bacter~
## 5 ASV_23~ 4          4          4          4          8      7.87 Bacter~
## 6 ASV_24~ 4          4          4          4          7      6.88 Bacter~
## 7 ASV_22~ 4          4          4          4          7      6.81 Bacter~
## 8 ASV_29~ 4          4          4          4          6      5.77 Bacter~
## 9 ASV_29~ 4          4          4          4          4      3.79 Bacter~
## 10 ASV_746 4          4          4          4          1      0.945 Bacter~
## 11 ASV_19~ 4          4          4          4          1      0.953 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 33 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.997      0.997 Positive
## 2      1      3 1.000      1.000 Positive
## 3      2      3 0.998      0.998 Positive
## # i 30 more rows
##

```

```
## $`6`
## # A tbl_graph: 11 nodes and 24 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 11 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_116 6          6          6          6          8          7.67 Bacter~
## 2 ASV_661 6          6          6          6          8          7.71 Bacter~
## 3 ASV_25~ 6          6          6          6          6          5.79 Bacter~
## 4 ASV_975 6          6          6          6          5          4.78 Bacter~
## 5 ASV_10~ 6          6          6          6          5          4.77 Bacter~
## 6 ASV_20~ 6          6          6          6          5          4.79 Bacter~
## 7 ASV_27~ 6          6          6          6          4          3.84 Bacter~
## 8 ASV_94 6          6          6          6          3          2.84 Bacter~
## 9 ASV_20~ 6          6          6          6          2          1.92 Bacter~
## 10 ASV_442 6          6          6          6          1          0.946 Bacter~
## 11 ASV_408 6          6          6          6          1          0.959 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 24 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2 0.974      0.974 Positive
## 2     1     3 0.966      0.966 Positive
## 3     2     3 0.979      0.979 Positive
## # i 21 more rows
##
## $`25`
## # A tbl_graph: 8 nodes and 22 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 8 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_1505 25          25          25          25          7          6.66 Bacter~
## 2 ASV_415 25          25          25          25          6          5.78 Bacter~
## 3 ASV_517 25          25          25          25          6          5.84 Bacter~
## 4 ASV_842 25          25          25          25          6          5.80 Bacter~
## 5 ASV_1719 25          25          25          25          6          5.85 Bacter~
## 6 ASV_2286 25          25          25          25          6          5.81 Bacter~
## 7 ASV_2637 25          25          25          25          5          4.78 Bacter~
## 8 ASV_1474 25          25          25          25          2          1.90 Bacter~
```

```

## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 22 x 5
##   from to weight correlation corr_direction
##   <int> <int> <dbl> <dbl> <chr>
## 1     1     2 0.947      0.947 Positive
## 2     1     3 0.951      0.951 Positive
## 3     2     3 0.970      0.970 Positive
## # i 19 more rows
##
## $`7`
## # A tbl_graph: 6 nodes and 5 edges
## #
## # An unrooted tree
## #
## # Node Data: 6 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_1896 7          7          7          7          3      2.87 Bacter~
## 2 ASV_454  7          7          7          7          2      1.89 Bacter~
## 3 ASV_1090 7          7          7          7          2      1.89 Bacter~
## 4 ASV_231  7          7          7          7          1      0.968 Bacter~
## 5 ASV_1800 7          7          7          7          1      0.943 Bacter~
## 6 ASV_2038 7          7          7          7          1      0.956 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 5 x 5
##   from to weight correlation corr_direction
##   <int> <int> <dbl> <dbl> <chr>
## 1     1     2 0.945      0.945 Positive
## 2     2     3 0.944      0.944 Positive
## 3     1     4 0.968      0.968 Positive
## # i 2 more rows
##
## $`12`
## # A tbl_graph: 5 nodes and 5 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 5 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_1148 12          12          12          12          4      3.85 Bacter~
## 2 ASV_1104 12          12          12          12          2      1.92 Bacter~

```

```

## 3 ASV_1854 12      12      12      12      2      1.95 Bacter~
## 4 ASV_115  12      12      12      12      1      0.955 Bacter~
## 5 ASV_2484 12      12      12      12      1      0.948 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 5 x 5
##   from   to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.959      0.959 Positive
## 2     1     3  0.990      0.990 Positive
## 3     2     3  0.957      0.957 Positive
## # i 2 more rows
##
## $`22`
## # A tbl_graph: 5 nodes and 7 edges
## #
## # An undirected simple graph with 1 component
## #
## # Node Data: 5 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>
## 1 ASV_1931 22          22          22          22          4        3.88 Bacter~
## 2 ASV_1320 22          22          22          22          3        2.89 Bacter~
## 3 ASV_1906 22          22          22          22          3        2.91 Bacter~
## 4 ASV_2704 22          22          22          22          3        2.91 Bacter~
## 5 ASV_1495 22          22          22          22          1        0.970 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 7 x 5
##   from   to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     1     2  0.971      0.971 Positive
## 2     1     3  0.969      0.969 Positive
## 3     2     3  0.959      0.959 Positive
## # i 4 more rows
##
## $`10`
## # A tbl_graph: 4 nodes and 3 edges
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>      <fct>      <ord>      <chr>      <ord>      <dbl>      <dbl> <chr>

```

```

## 1 ASV_2059 10      10      10      10      2      1.89 Bacter~
## 2 ASV_1965 10      10      10      10      2      1.91 Bacter~
## 3 ASV_544  10      10      10      10      1      0.964 Bacter~
## 4 ASV_1607 10      10      10      10      1      0.946 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.943      0.943 Positive
## 2      2      3 0.964      0.964 Positive
## 3      1      4 0.946      0.946 Positive
##
## $`11`
## # A tbl_graph: 4 nodes and 3 edges
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_614  11          11          11          11          2      1.91 Bacter~
## 2 ASV_2916 11          11          11          11          2      1.93 Bacter~
## 3 ASV_65    11          11          11          11          1      0.943 Archaea
## 4 ASV_2168 11          11          11          11          1      0.965 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
##   from    to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1      1      2 0.966      0.966 Positive
## 2      1      3 0.943      0.943 Positive
## 3      2      4 0.965      0.965 Positive
##
## $`15`
## # A tbl_graph: 4 nodes and 3 edges
## #
## # An unrooted tree
## #
## # Node Data: 4 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>     <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_172  15          15          15          15          2      1.92 Bacter~
## 2 ASV_246  15          15          15          15          2      1.91 Bacter~

```

```

## 3 ASV_43 15 15 15 15 1 0.969 Bacter~
## 4 ASV_2211 15 15 15 15 1 0.955 Bacter~
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 3 x 5
## from to weight correlation corr_direction
## <int> <int> <dbl> <dbl> <chr>
## 1 1 2 0.952 0.952 Positive
## 2 1 3 0.969 0.969 Positive
## 3 2 4 0.955 0.955 Positive
##
## $Others
## # A tbl_graph: 39 nodes and 34 edges
## #
## # An undirected simple graph with 14 components
## #
## # Node Data: 39 x 14 (active)
## name modularity modularity2 modularity3 Modularity Degree Strength Kingdom
## <chr> <fct> <ord> <chr> <ord> <dbl> <dbl> <chr>
## 1 ASV_316 28 Others Others Others 3 2.89 Bacter~
## 2 ASV_610 28 Others Others Others 3 2.94 Bacter~
## 3 ASV_11~ 29 Others Others Others 3 2.94 Bacter~
## 4 ASV_18~ 29 Others Others Others 3 2.95 Bacter~
## 5 ASV_11~ 27 Others Others Others 3 2.91 Bacter~
## 6 ASV_18~ 28 Others Others Others 3 2.91 Bacter~
## 7 ASV_20~ 29 Others Others Others 3 2.89 Bacter~
## 8 ASV_24~ 27 Others Others Others 3 2.89 Bacter~
## 9 ASV_27~ 29 Others Others Others 3 2.95 Bacter~
## 10 ASV_28~ 28 Others Others Others 3 2.90 Bacter~
## # i 29 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 34 x 5
## from to weight correlation corr_direction
## <int> <int> <dbl> <dbl> <chr>
## 1 1 2 0.971 0.971 Positive
## 2 3 4 0.987 0.987 Positive
## 3 1 6 0.958 0.958 Positive
## # i 31 more rows

```

```
module5_2_8_subgraph_info$sub_graph_select
```

```
## # A tbl_graph: 77 nodes and 582 edges
```

```
## #
## # An undirected simple graph with 3 components
## #
## # Node Data: 77 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <fct>      <ord>      <chr>      <ord>      <dbl>    <dbl> <chr>
## 1 ASV_649 5          5          5          5          27      26.5 Bacter~
## 2 ASV_705 5          5          5          5          27      26.5 Bacter~
## 3 ASV_12~ 5          5          5          5          27      26.5 Bacter~
## 4 ASV_13~ 5          5          5          5          27      26.5 Bacter~
## 5 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 6 ASV_14~ 5          5          5          5          27      26.5 Bacter~
## 7 ASV_24~ 5          5          5          5          27      26.5 Bacter~
## 8 ASV_25~ 5          5          5          5          27      26.4 Bacter~
## 9 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## 10 ASV_28~ 5          5          5          5          27      26.5 Bacter~
## # i 67 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## #   Genus <chr>, Species <chr>
## #
## # Edge Data: 582 x 5
##   from      to weight correlation corr_direction
##   <int> <int> <dbl>      <dbl> <chr>
## 1     56    74 0.952      0.952 Positive
## 2     57    74 0.954      0.954 Positive
## 3     58    74 0.947      0.947 Positive
## # i 579 more rows
```

5.2 Extract subgraph by sample

Chapter 6

Network layout

Load R Package

```
library(tidyverse)
library(ggNetView)
```

Example data

```
# Access built-in example datasets in ggNetView

# Relative abundance table of rarefied ASVs or OTUs
data("otu_rare_relative")
dim(otu_rare_relative)
```

```
## [1] 2859  18
```

```
otu_rare_relative[1:5, 1:5]
```

```
##           K01           K02           K03           K04           K05
## ASV_1 0.03306667 0.05453333 0.02013333 0.03613333 0.02686667
## ASV_2 0.05750000 0.03393333 0.06046667 0.05810000 0.07320000
## ASV_3 0.01733333 0.01296667 0.02290000 0.02336667 0.03106667
## ASV_4 0.04266667 0.01093333 0.01416667 0.01933333 0.03346667
## ASV_6 0.02646667 0.01856667 0.02110000 0.02353333 0.03806667
```

```
# Taxonomic annotation table of ASVs or OTUs
data("tax_tab")
dim(tax_tab)
```

```
## [1] 2859      8
```

```
tax_tab[1:5, 1:5]
```

```
## # A tibble: 5 x 5
##   OTUID Kingdom Phylum      Class      Order
##   <chr>  <chr>    <chr>      <chr>      <chr>
## 1 ASV_2  Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 2 ASV_3  Bacteria Verrucomicrobia Spartobacteria Unassigned
## 3 ASV_31 Bacteria Actinobacteria Actinobacteria Actinomycetales
## 4 ASV_27 Archaea  Thaumarchaeota  Unassigned  Nitrososphaerales
## 5 ASV_9  Bacteria Unassigned      Unassigned  Unassigned
```

Build graph object

```
graph_obj <- build_graph_from_mat(
  mat = otu_rare_relative,
  transform.method = "none", # based your input data
  r.threshold = 0.7,
  p.threshold = 0.05,
  method = "WGCNA",
  cor.method = "pearson",
  proc = "BH",
  module.method = "Fast_greedy",
  node_annotation = tax_tab,
  top_modules = 15,
  seed = 1115
)

graph_obj
```

```
## # A tbl_graph: 2049 nodes and 9602 edges
## #
## # An undirected simple graph with 100 components
## #
## # Node Data: 2,049 x 14 (active)
##   name      modularity modularity2 modularity3 Modularity Degree Strength Kingdom
##   <chr>    <dbl>      <dbl>      <dbl>      <dbl>    <dbl>    <dbl>    <chr>
## 1 ASV_916 1          1          1          1          58      50.5 Bacter~
## 2 ASV_777 1          1          1          1          58      48.7 Bacter~
## 3 ASV_606 1          1          1          1          55      45.8 Bacter~
## 4 ASV_740 1          1          1          1          54      47.2 Bacter~
## 5 ASV_14~ 1          1          1          1          54      44.5 Bacter~
## 6 ASV_23~ 1          1          1          1          54      47.4 Bacter~
```

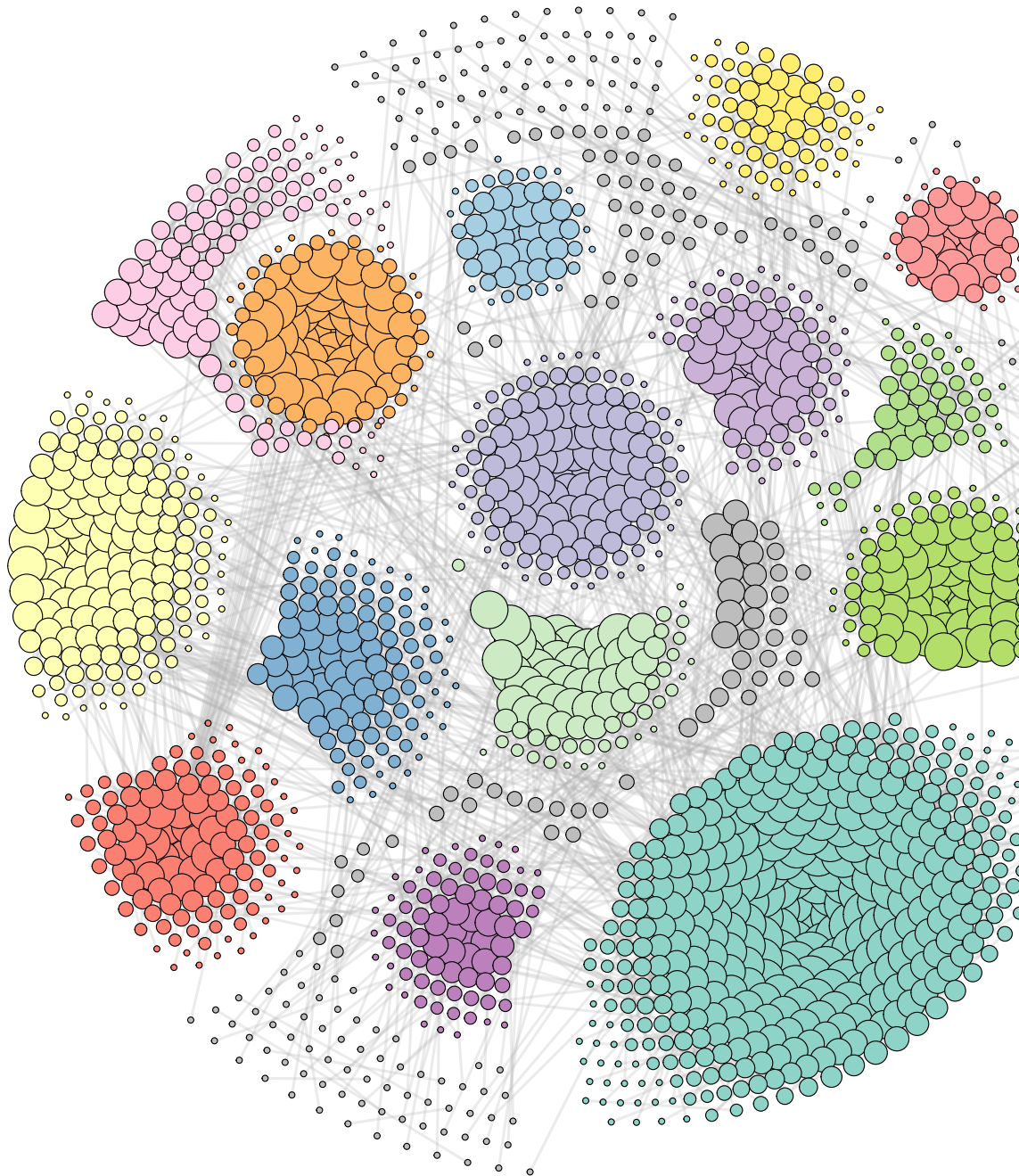
```
## 7 ASV_15~ 1 1 1 1 52 45.3 Bacter~
## 8 ASV_24~ 1 1 1 1 52 43.0 Bacter~
## 9 ASV_19~ 1 1 1 1 52 43.0 Bacter~
## 10 ASV_568 1 1 1 1 51 45.1 Bacter~
## # i 2,039 more rows
## # i 6 more variables: Phylum <chr>, Class <chr>, Order <chr>, Family <chr>,
## # Genus <chr>, Species <chr>
## #
## # Edge Data: 9,602 x 5
## from to weight correlation corr_direction
## <int> <int> <dbl> <dbl> <chr>
## 1 1771 1825 0.793 0.793 Positive
## 2 594 597 0.895 0.895 Positive
## 3 588 597 0.864 0.864 Positive
## # i 9,599 more rows
```

6.1 Gephi layout

```
p <- ggNetView(
  graph_obj = graph_obj,
  layout = "gephi",
  center = F,
  shrink = 0.8,
  layout.module = "random",
  group.by = "Modularity",
  fill.by = "Modularity"
)

p
```

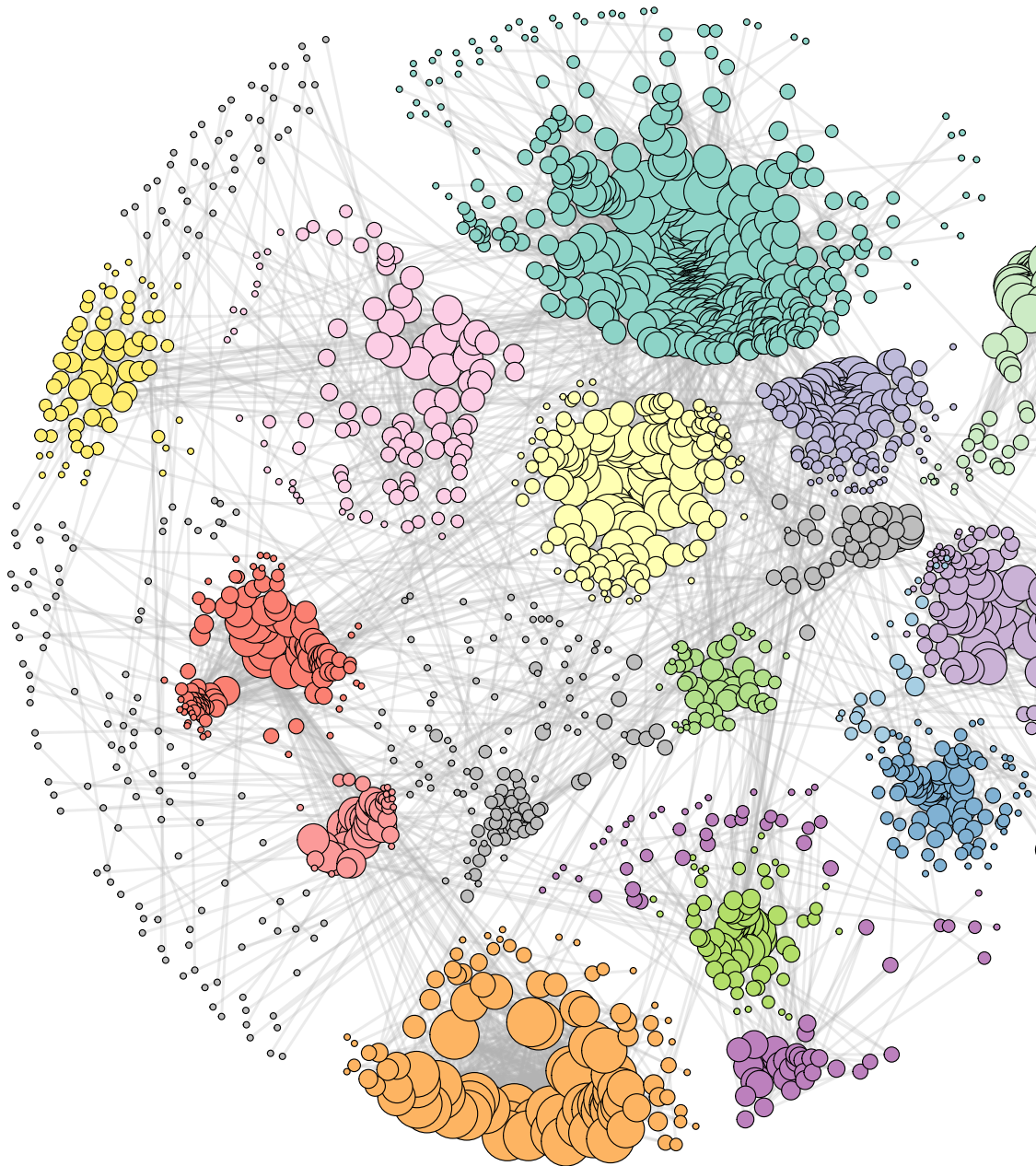
Node = 2049
Edge = 9602



6.2 Fruchterman–Reingold force-directed layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "fr2",  
  center = F,  
  shrink = 0.8,  
  layout.module = "random",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

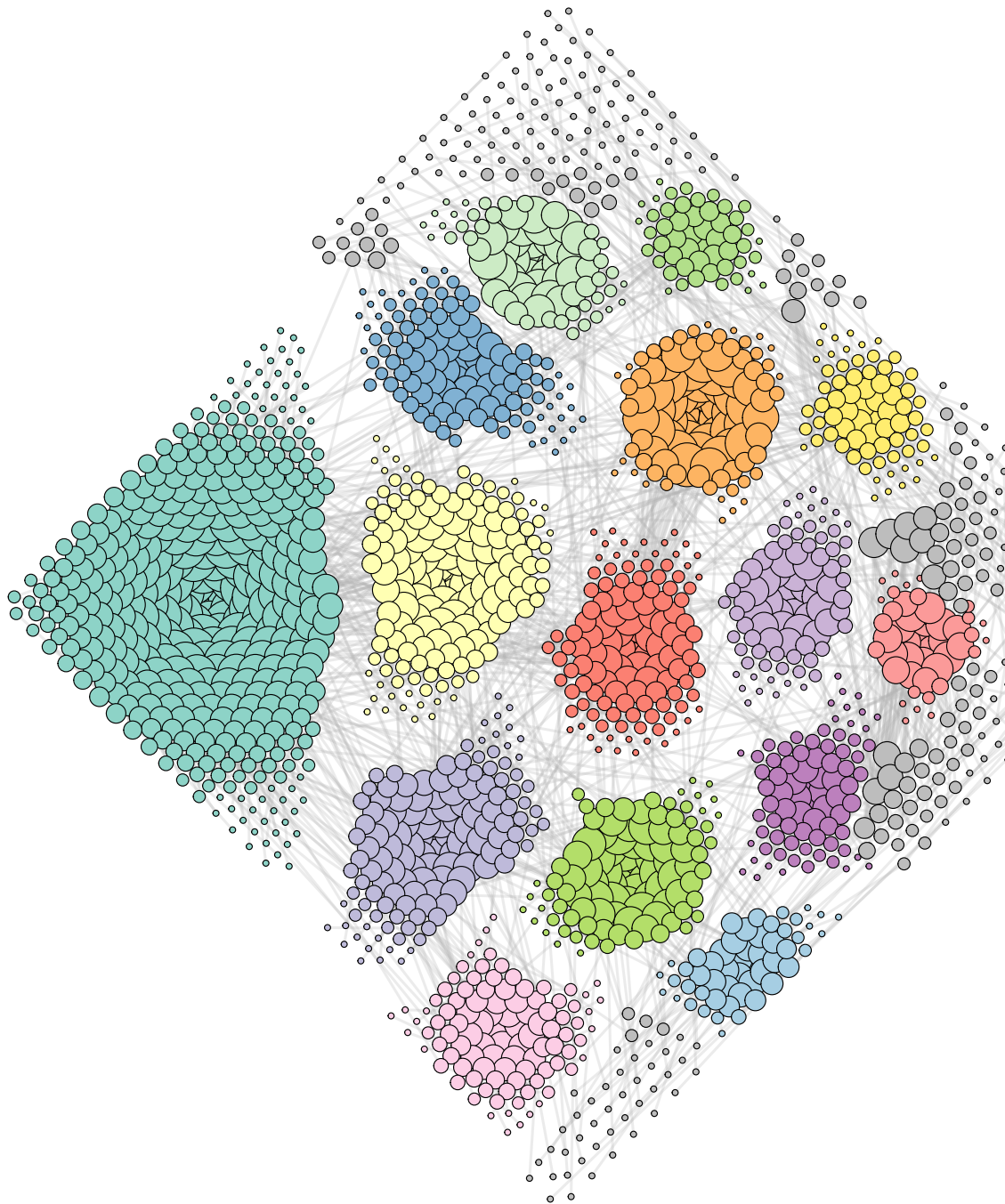
Node = 2049
Edge = 9602



6.3 Diamond layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "diamond",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

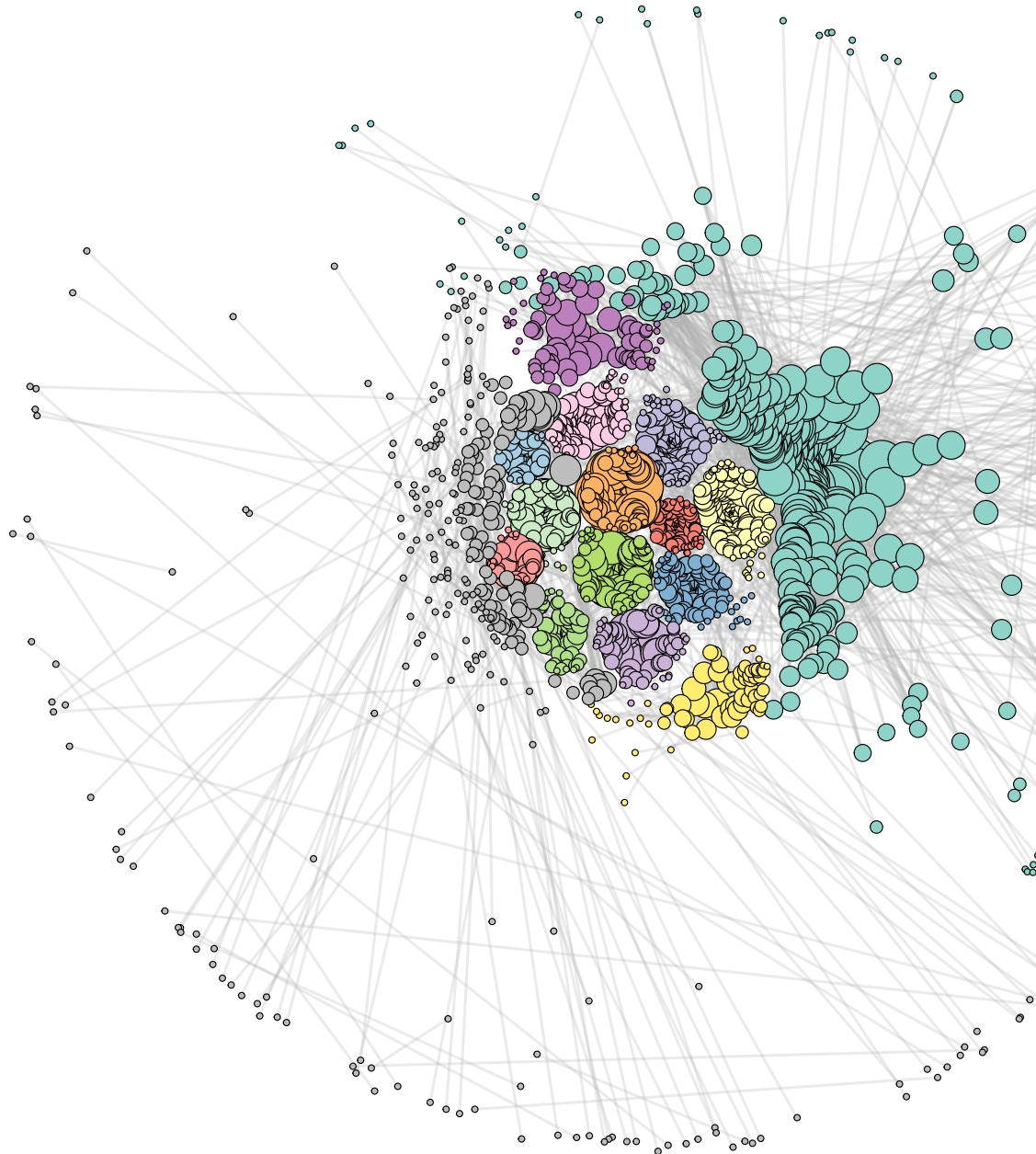

Node = 2049
Edge = 9602



6.4 Kk layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "kk",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

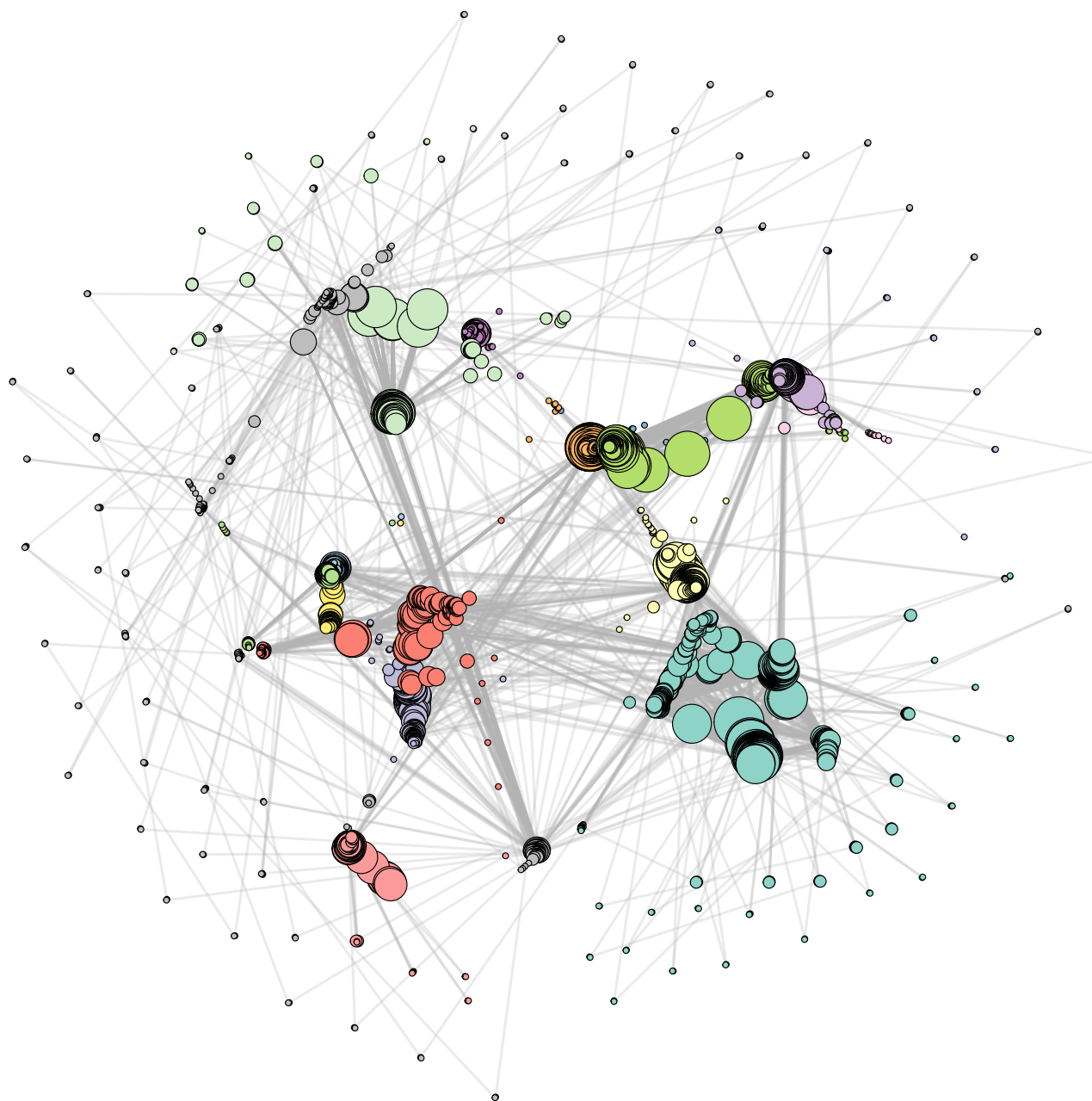
Node = 2049
Edge = 9602



6.5 Nicley layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "nicely",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 12; retrying with k_nn = 32.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 32; retrying with k_nn = 52.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 52; retrying with k_nn = 72.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 72; retrying with k_nn = 92.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 92; retrying with k_nn = 115.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 115; retrying with k_nn = 144.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 144; retrying with k_nn = 180.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 180; retrying with k_nn = 225.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 225; retrying with k_nn = 282.  
  
## Warning in ggNetView(graph_obj = graph_obj, layout = "nicely", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 282; retrying with k_nn = 353.
```

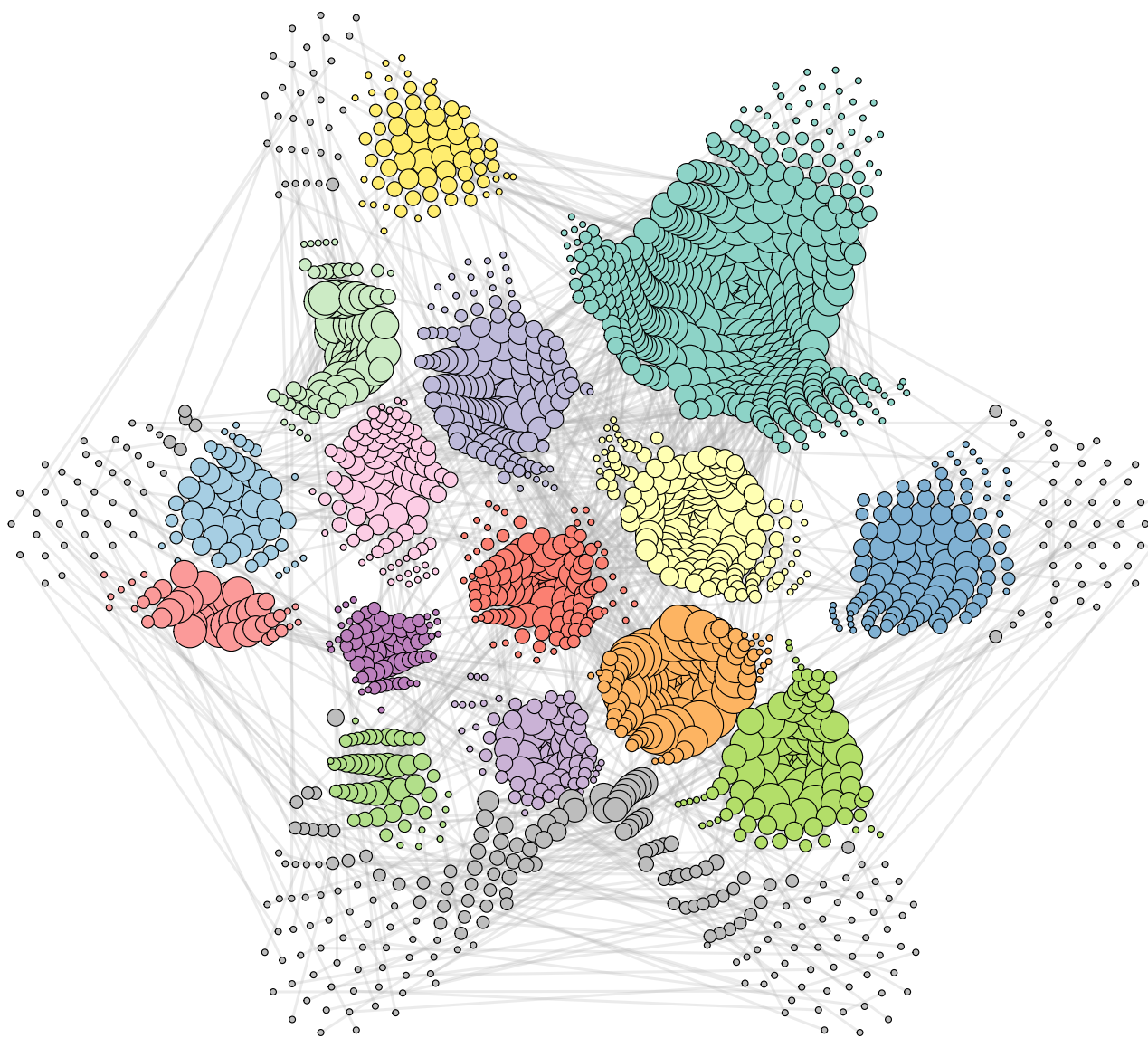

Node = 2049
Edge = 9602



6.6 Petal layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "petal",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

Node = 2049
Edge = 9602

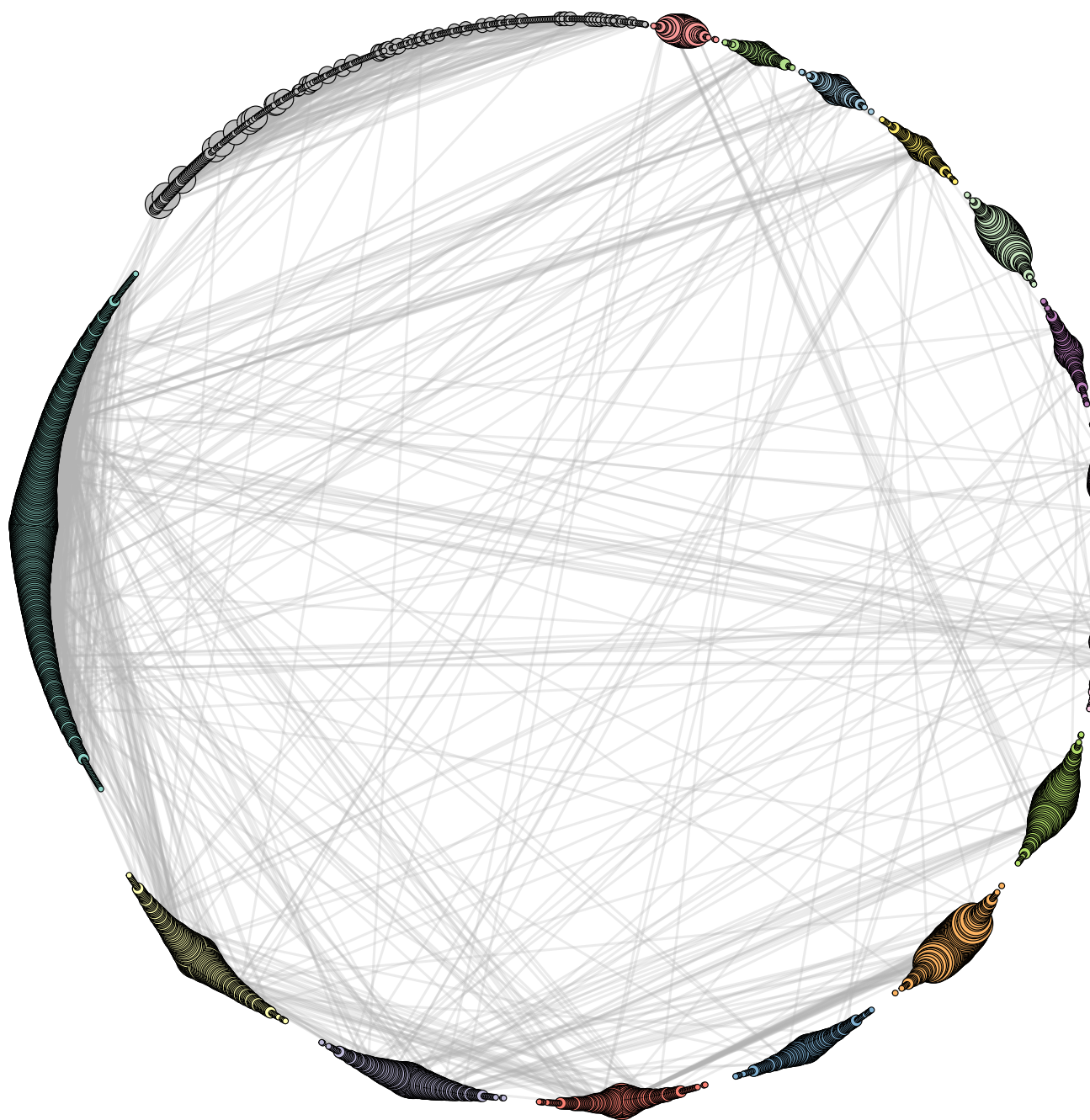


6.7 Circle layout

6.8 Petal layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "circle",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

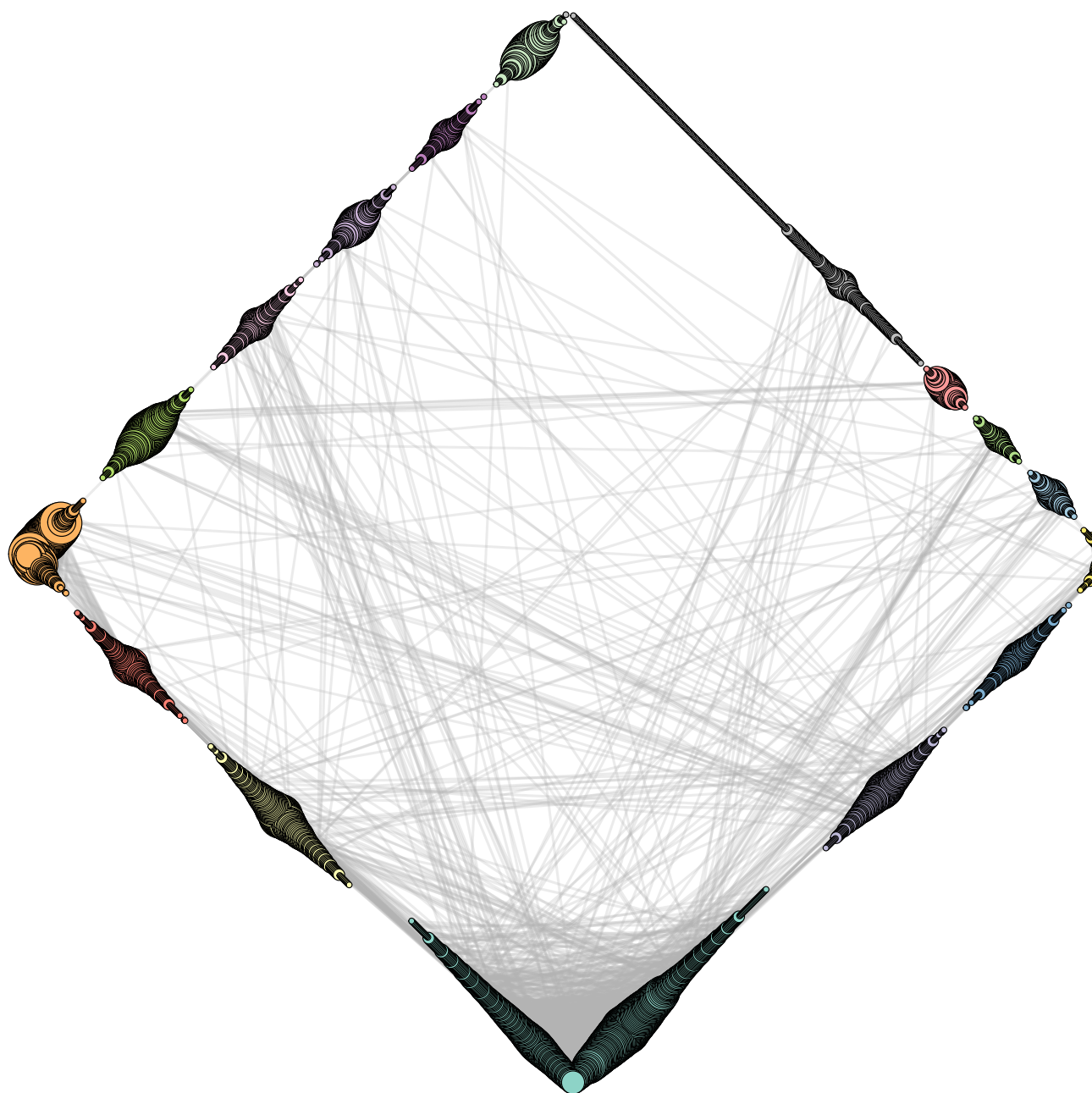

Node = 2049
Edge = 9602



6.9 Diamond outline layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "diamond_outline",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

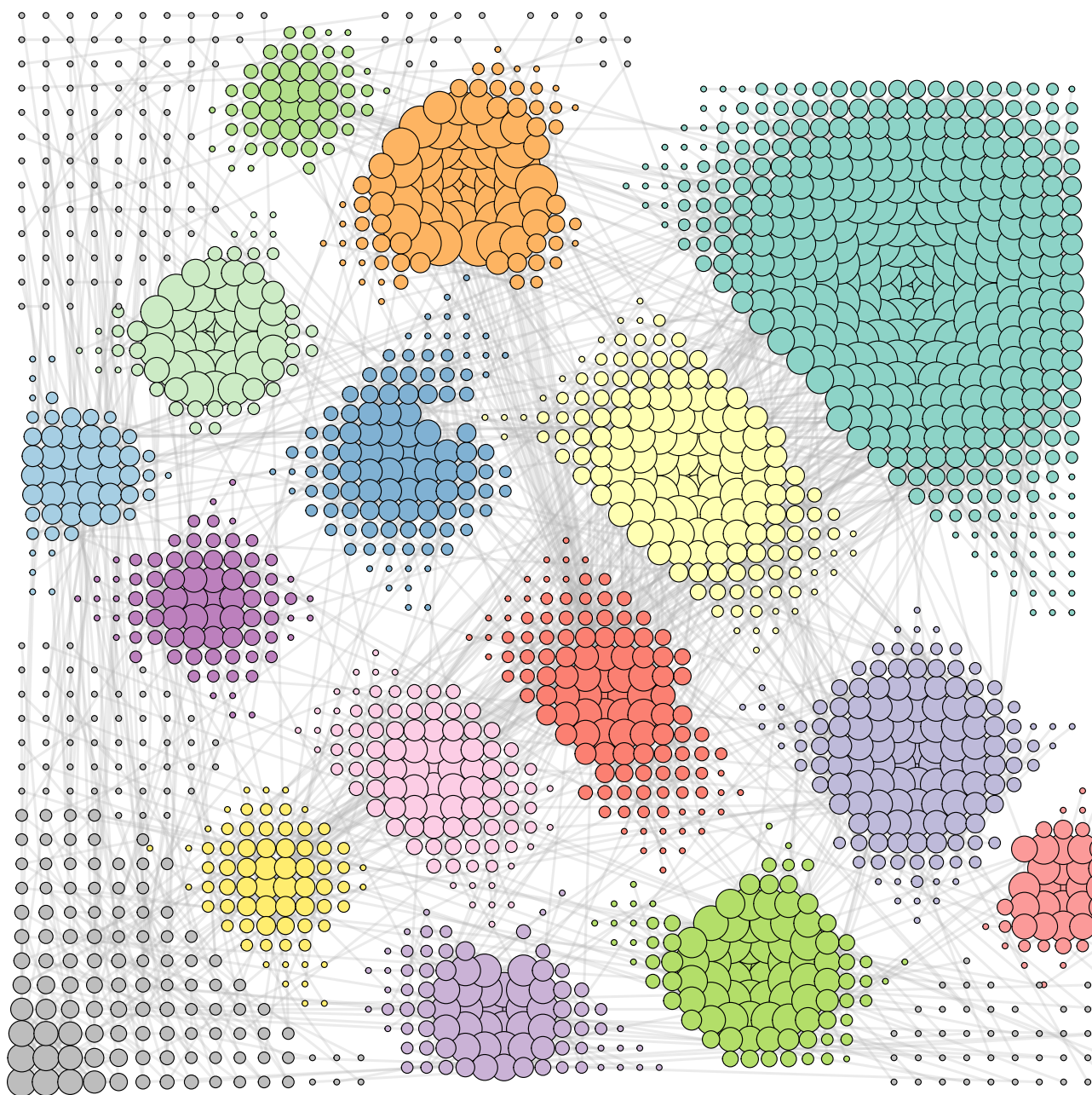
Node = 2049
Edge = 9602



6.10 Grid layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "grid",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

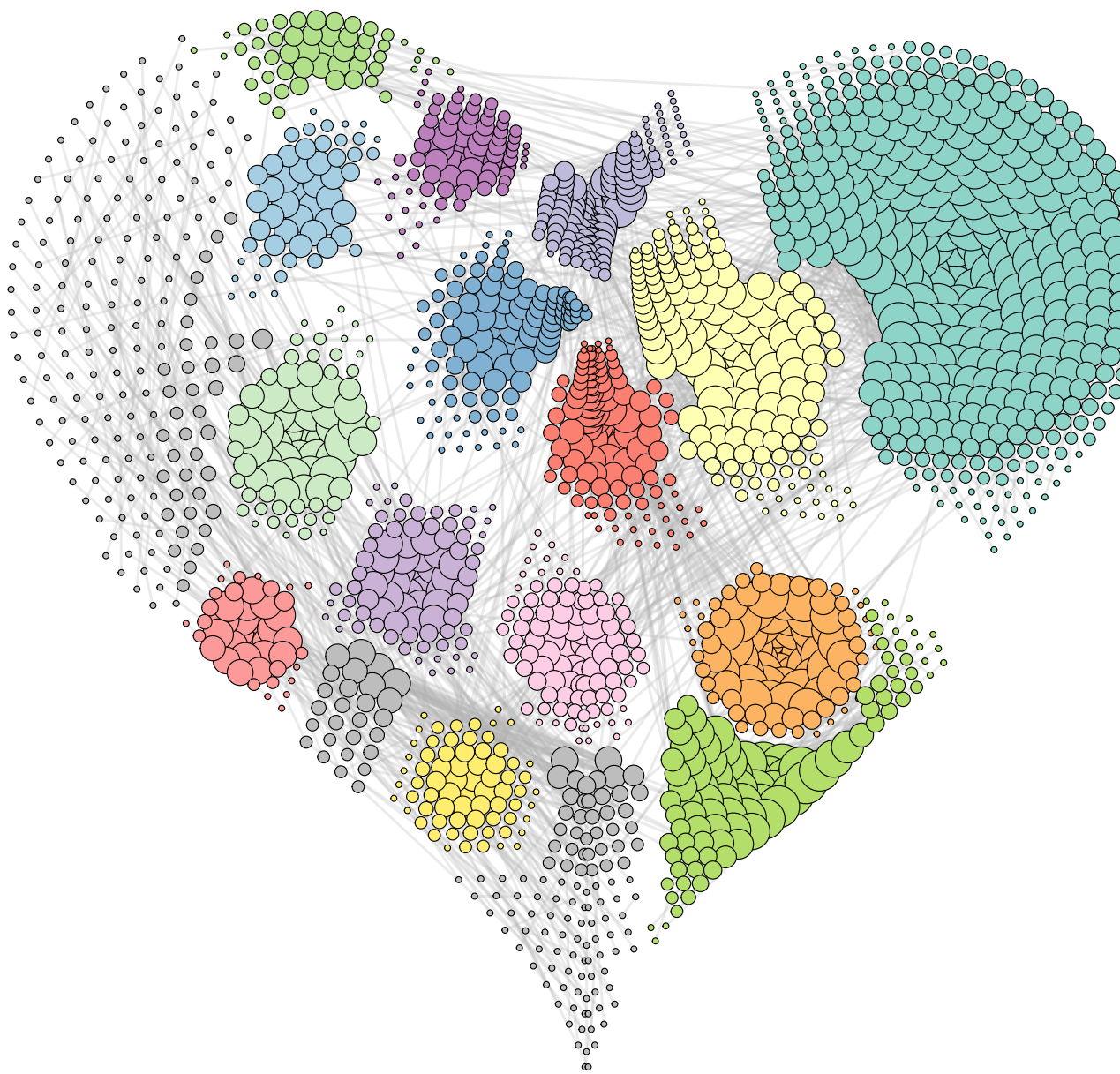
Node = 2049
Edge = 9602



6.11 Heart_centered layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "heart_centered",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```


Node = 2049
Edge = 9602



6.12 Lgl layout

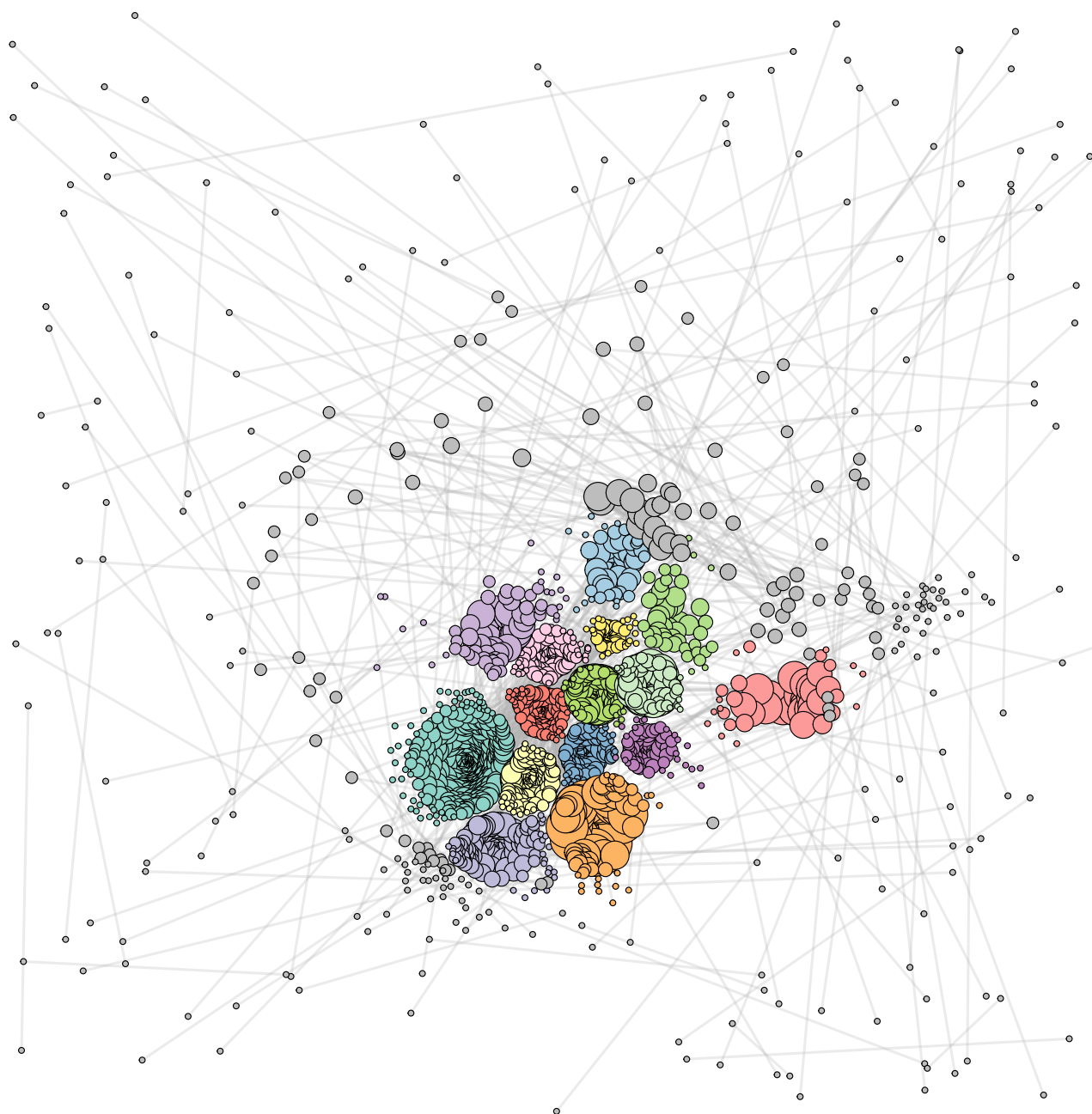
```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "lgl",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)
```

```
## Warning in alg_fun(graph): LGL layout does not support disconnected graphs yet.  
## Source: layout/large_graph.c:179
```

```
## Warning in ggNetView(graph_obj = graph_obj, layout = "lgl", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 12; retrying with k_nn = 32.
```

```
p
```

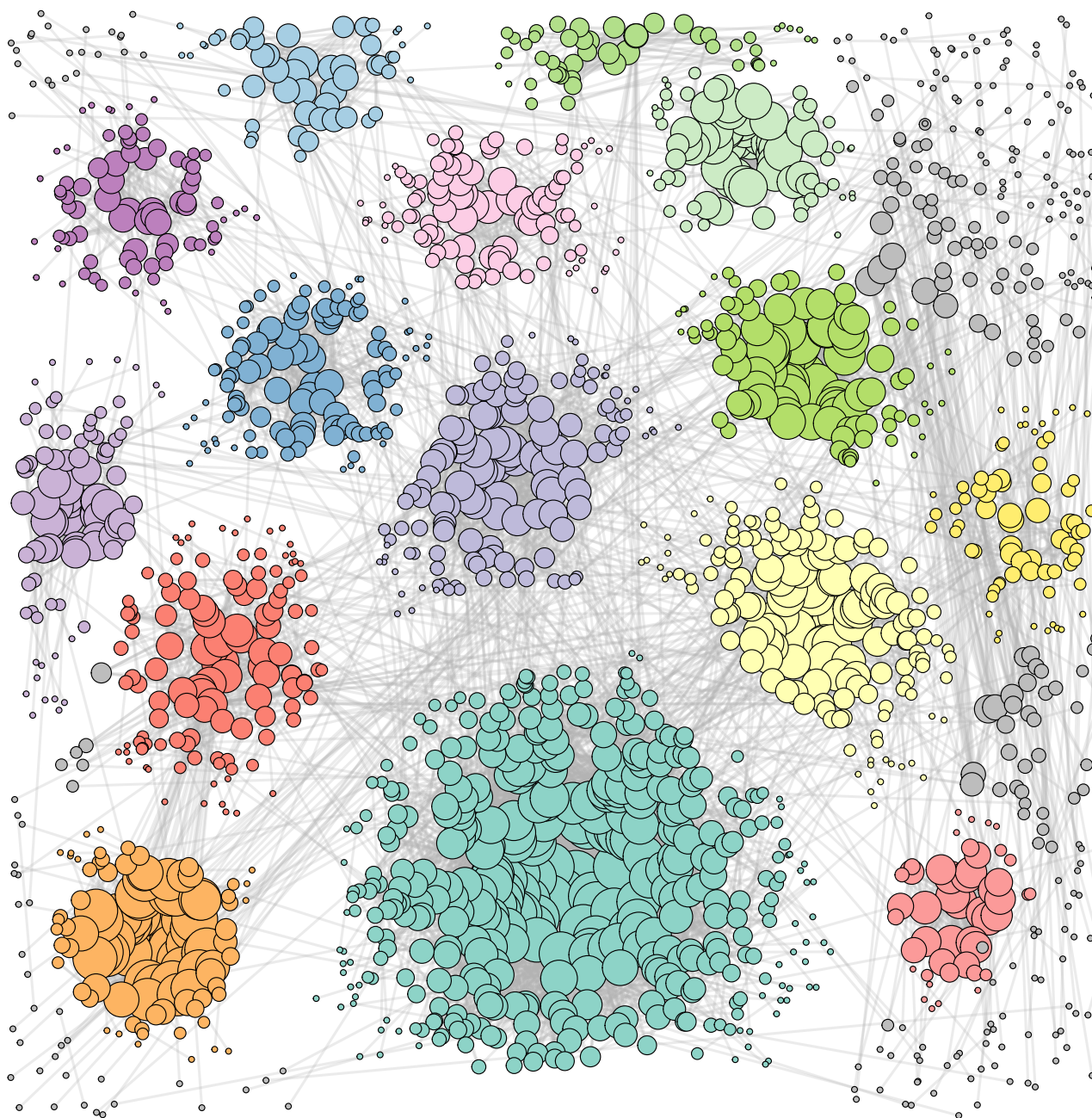

Node = 2049
Edge = 9602



6.13 Randomly layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "randomly",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

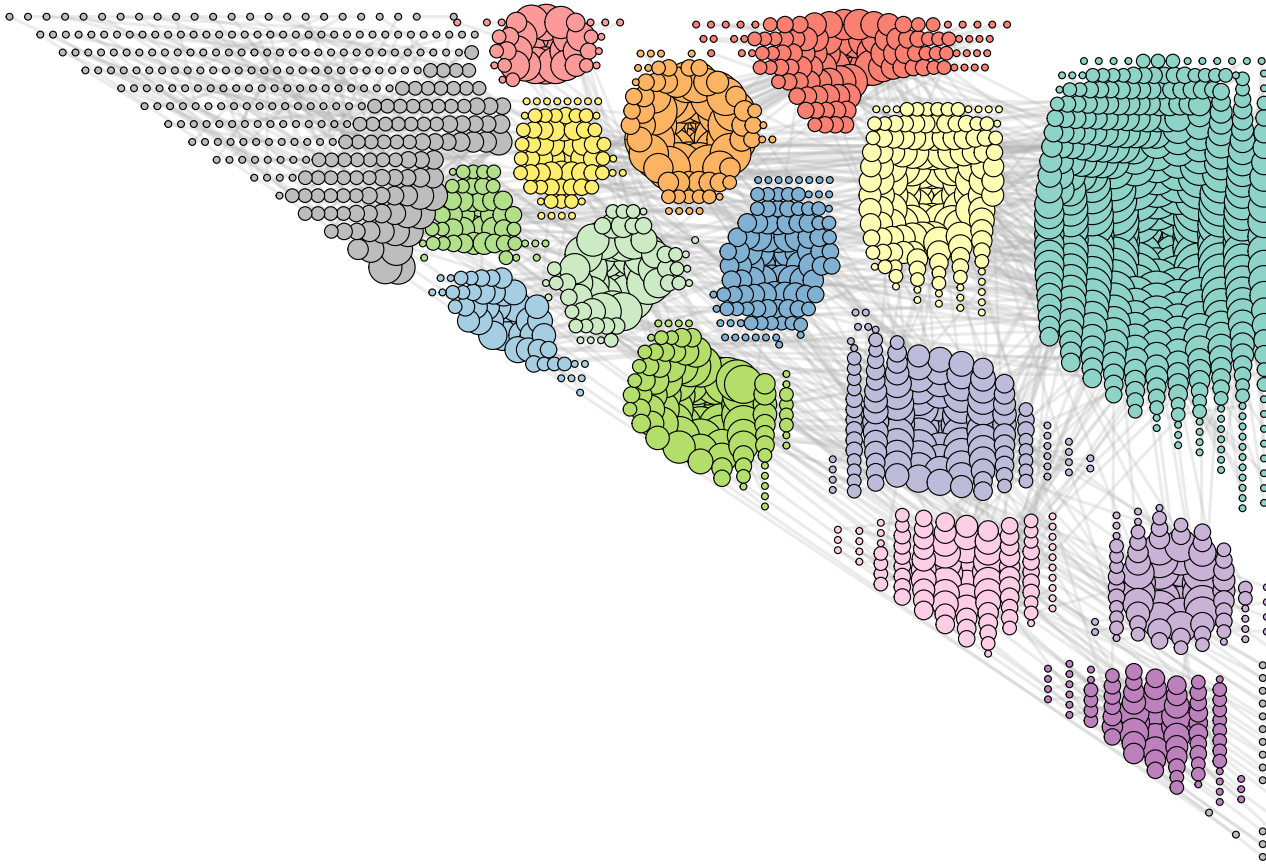
Node = 2049
Edge = 9602



6.14 Rectangle layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "rectangle",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

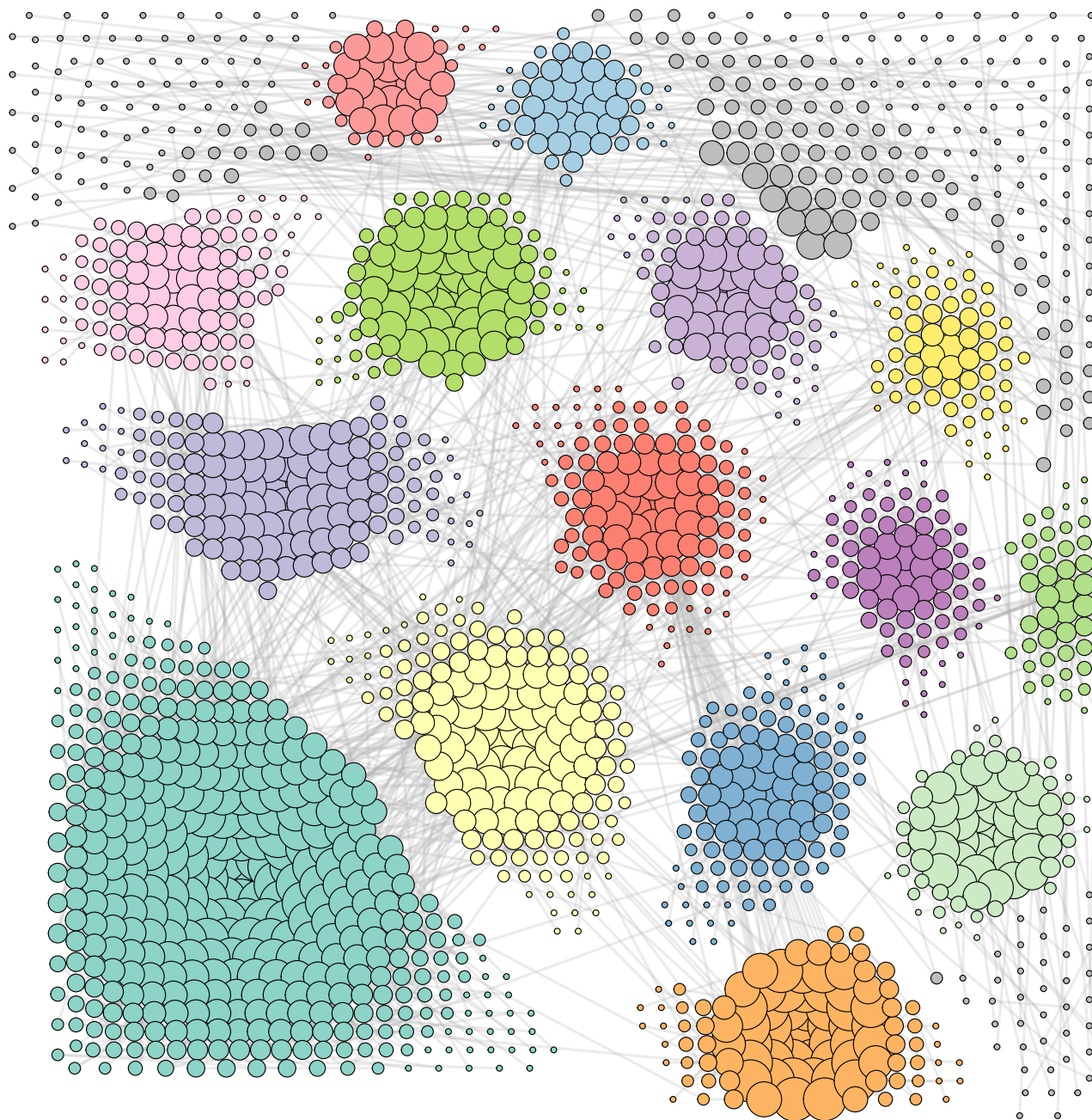
Node = 2049
Edge = 9602



6.15 Square layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "square",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

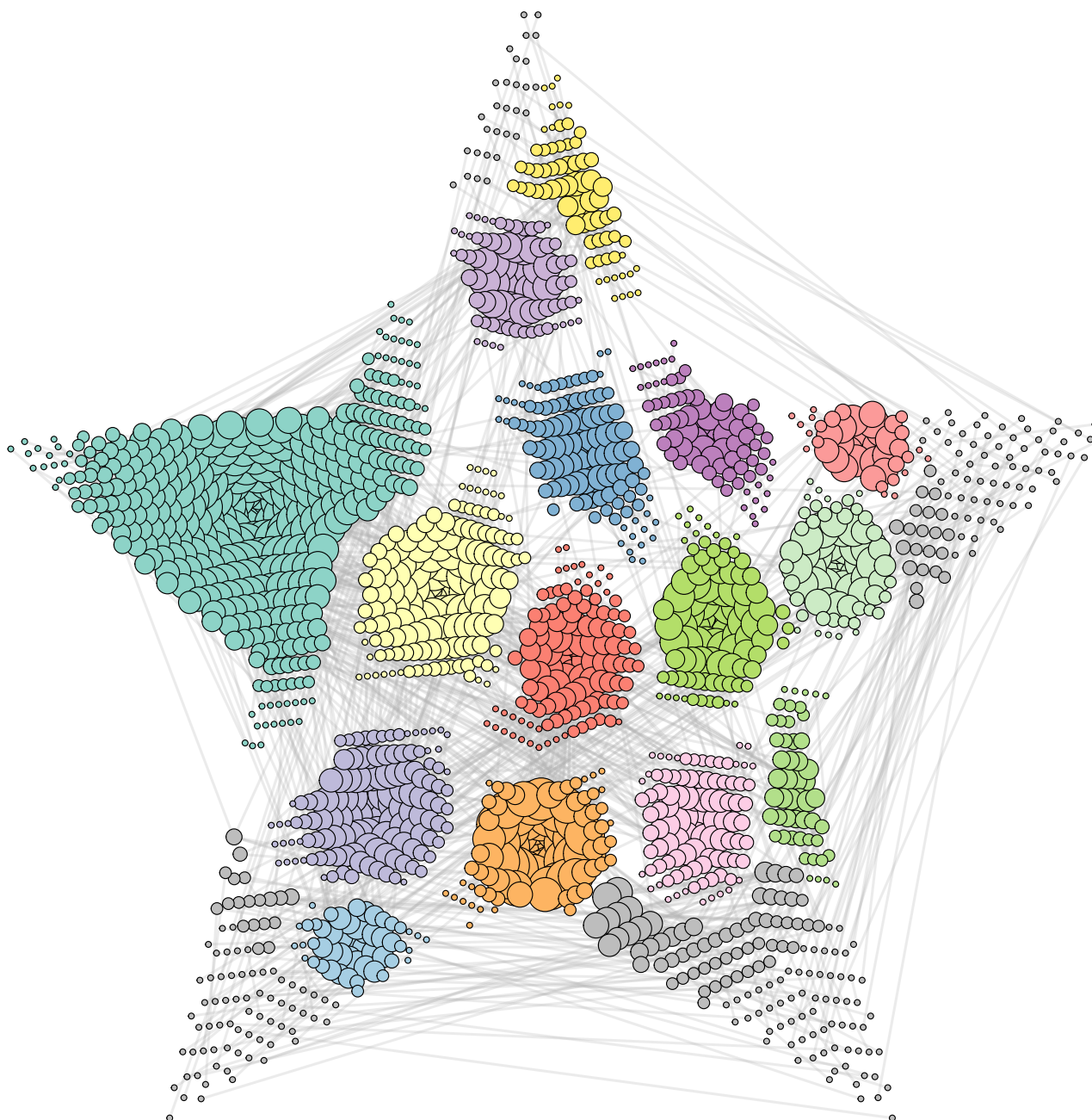

Node = 2049
Edge = 9602



6.16 Star layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "star",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

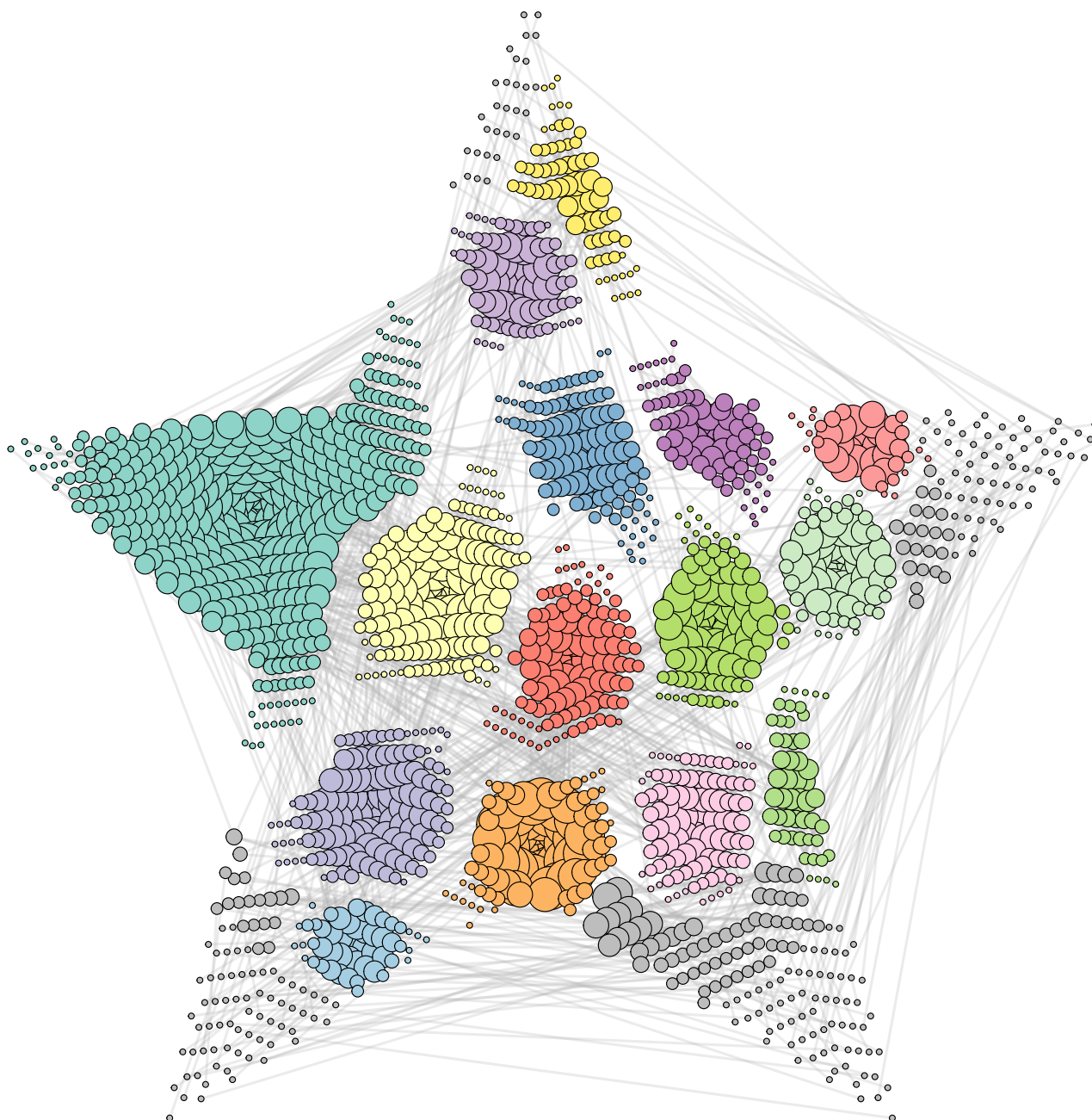

Node = 2049
Edge = 9602



6.17 Star__concentric layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "star_concentric",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)  
p
```

Node = 2049
Edge = 9602



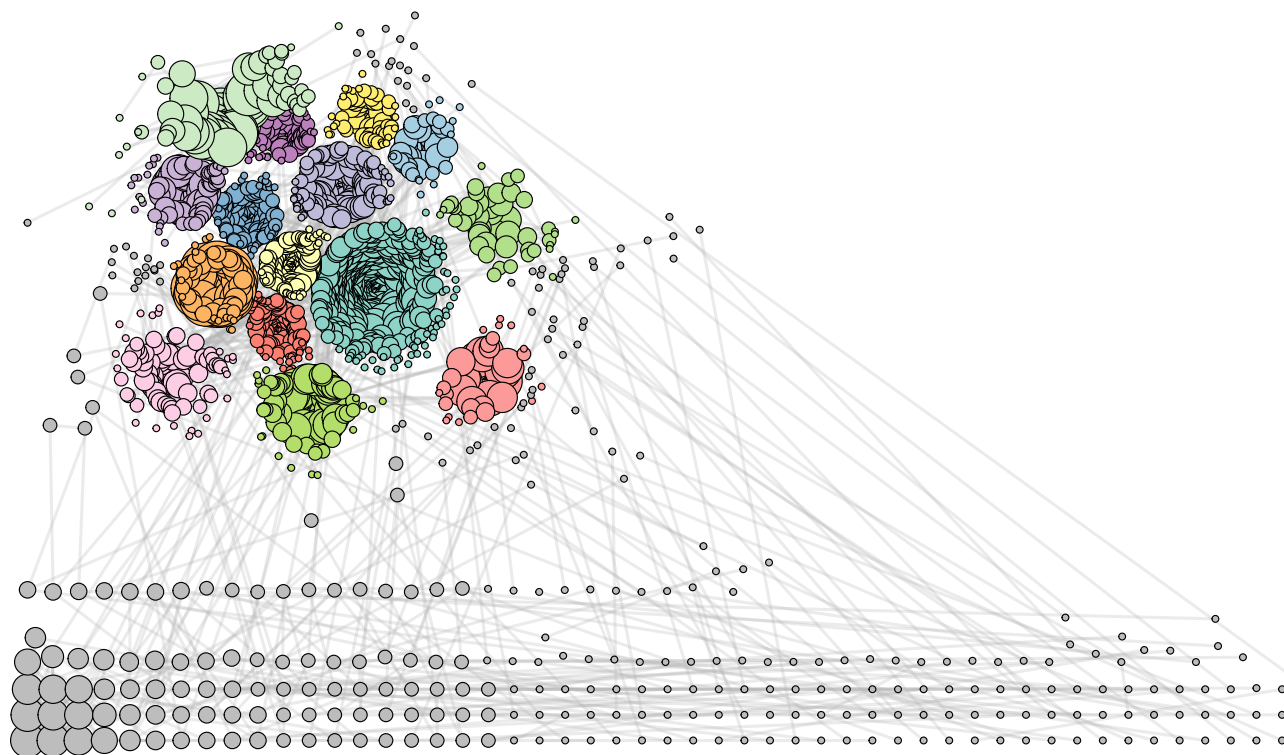
6.18 Stress layout

```
p <- ggNetView(  
  graph_obj = graph_obj,  
  layout = "stress",  
  center = F,  
  shrink = 0.8,  
  layout.module = "adjacent",  
  group.by = "Modularity",  
  fill.by = "Modularity"  
)
```

```
## Warning in ggNetView(graph_obj = graph_obj, layout = "stress", center = F, :  
## `layout.module = 'adjacent'` failed at k_nn = 12; retrying with k_nn = 32.
```

```
p
```

Node = 2049
Edge = 9602



Chapter 7

Network topology information

- 7.1 Get network topology information
- 7.2 Get network topology information with matrix
- 7.3 Get network topology information by parallel
- 7.4 Get network topology information with matrix by parallel

Chapter 8

Network comparison

8.1 Subgraph comparison

8.2 Comparison of multi-sample networks

Chapter 9

Network & Environment

9.1 Network Environment

Chapter 10

Multi-omics network analysis

10.1 Multi-omics